

Sparse indicator quadratics: exact complexity and smoothed separator messages

Abstract

We study positive-definite quadratic optimization with one binary indicator for each continuous variable. Independent finite-grid perturbations of the indicator penalties permit exact construction of every separator value function on prescribed bounded boxes at fixed treewidth. The output contains every support optimal anywhere, including at a single tie point. The algorithm is correct for every draw and has polynomial expected bit complexity in the input length, numerical coefficient and spectral bounds, box radius, and inverse perturbation scale. It needs logarithmically many random bits per coordinate and no deterministic support margin or graph-growth assumption. The proof combines a near-optimal support count, certified enumeration from restricted additive approximation oracles, and a parameter net; a finite-domain dynamic program supplies the oracle. Complementary lower bounds explain the restrictions. Exact optimization is NP-hard at bandwidth two with near-identity matrices. Fixed local coefficients can generate exponentially many indispensable scalar message formulas, although pruning admits matching power-law bounds. On uniformly conditioned stars, an inverse-principal polytope has a correlation-polytope face, and a specific exact decision-diagram state scheme has exponential width even after row normalization. These representation bounds are distinct from hardness of optimizing the original tree problem. Supporting results give all fixed moments of active-label counts, polynomial expected connected-region counts and exact labeled subdivisions at fixed dimension, sharper scalar and planar estimates, and direct recursive algorithms under diagonal dominance.

1 Introduction

An indicator quadratic problem chooses which continuous coordinates may be nonzero and then minimizes a convex quadratic objective on the selected coordinates. Once that choice is fixed, a linear system solves the continuous problem. The difficulty lies in selecting the indicators. Sparse interaction graphs suggest dynamic programming, but eliminating a subtree creates a function of its continuous boundary variables. A small separator controls the dimension of this function, not the number of supports that can define it.

We give conditions under which all these value functions can be constructed exactly in expected polynomial time. The matrix is positive definite and its support graph has fixed treewidth. We perturb only the indicator penalties, using independent uniform choices from a rational grid of linear size. For each prescribed bounded separator box, the algorithm returns a *dictionary*: a list of full quadratic functions whose pointwise minimum is the exact value function. It retains every support that attains this minimum anywhere, even at a single tie point. It may also retain supports that are never optimal. The construction does not partition the box into optimality regions.

The main contribution is this complete message construction under a shared penalty perturbation. It separates two questions that are often conflated: whether a short representation exists and whether one can find it. First, a short shattering argument bounds the expected number of supports near the

optimum at any fixed parameter. Second, a certified version of classical partition enumeration finds a controlled superset of those supports using only additive approximation. Finally, a deterministic parameter net captures every support that is optimal somewhere. Computing each message directly avoids a recursive product of child dictionaries. This is useful when reusable exact conditional value functions are required, as opposed to one optimal solution at one parameter.

Our complexity bound is polynomial in numerical bounds as well as input length. In particular, it depends on an upper spectral bound, the inverse lower spectral bound, a bound on the linear coefficients, the separator box radius, and the inverse perturbation scale. It is not a bound polynomial in the logarithms of all these quantities. The exact answer concerns the perturbed instance. We also obtain a certified additive interval for the unperturbed optimum.

Three complementary results explain why additional conditions matter. Section 5 proves exact NP-hardness at bandwidth two with positive penalties and matrices arbitrarily close to the identity. It separates large-coefficient constant-gap hardness from bounded-coefficient exact hardness, and shows why bounded penalty noise cannot erase arbitrary coefficient magnitudes. Section 6 constructs a scalar message with exponentially many necessary formulas from a fixed coefficient alphabet, and proves a matching power law for approximation by pruning its original quadratics. Appendix A gives two limits for universal representations on stars: a correlation-polytope face in the inverse-principal polytope and exponential width for exact projected-row decision diagrams, including row normalization. These results distinguish the complexity of optimization, conditional value functions, and particular universal representations.

The supporting appendices separate three output notions: all active support labels, distinct necessary polynomial formulas, and connected regions of a unique winning label. Appendix B proves every fixed moment of active-label counts, including atomic noise, and combines those moments with classical algebraic bounds. This gives polynomial expected connected region counts in every fixed dimension and an exact subdivision labeled with all minimizing supports on every cell, using a larger polynomial-size noise grid than the main dictionary theorem. Appendix C gives sharper continuous scalar and planar estimates and a discrepancy transfer for necessary formulas. Appendix D proves direct block and fixed-width recursions under diagonal dominance. These derived results retain their distinct assumptions and outputs; the general geometric, coarea, and CAD tools are established.

1.1 Relation to prior work

Bhathena, Fattahi, Gómez, and Küçükyavuz [8] already solve the unperturbed problem on arbitrary trees in quadratic arithmetic time and memory. Their parametric value functions and conjugate calculations are direct antecedents. Our result does not improve that tree-specific guarantee. Their subsequent work on structured graphs [7, Definition 5, Lemma 9, and Theorem 1] constructs exact parametric dynamic programs with complexity controlled by treewidth, volume growth, conditioning, and a deterministic bound on the multiplicity of nearly optimal local supports. We instead control that multiplicity in expectation under penalty perturbations. Fixed treewidth suffices without a volume-growth or degree restriction, but we solve a perturbed problem and give no comparable practical running-time claim.

Approximation on sparse graphs is also established. Bienstock and Muñoz [10] develop bounded-treewidth formulations for polynomial optimization. Bienstock and Chen [9, Theorem 1.4] treat structured indicator quadratic models, with a general guarantee allowing controlled violation of certain constraints. Gómez, Han, and Lozano [19] develop decision-diagram approximations for banded quadratics. Choi et al. [13, Sections 7–8] provide exact structured diagrams and approximation results using decay, volume growth, and boundary-size bounds for the support graph of a matrix or its inverse. Our proof uses an elementary feasible grid approximation on a tree

decomposition; we do not claim a new approximation principle or a uniformly stronger result than these methods.

The probabilistic and enumeration ingredients have close classical antecedents. The sandwich inequality for shattered sets is established; we use the form stated by Kozma and Moran [21, Theorem 5]. Lawler’s partition method [23] lists ranked solutions from an exact optimization oracle. Our additive-oracle version gives a two-sided near-optimality inclusion instead of an exact ranking. Röglin and Teng [28, Sections 6.1–6.2] prove higher winner-gap bounds and expected polynomial smoothed optimization from pseudopolynomial algorithms. Dughmi and Roughgarden [16, Proposition 2.4] explicitly state a conversion from an approximation scheme to exact smoothed optimization for binary linear objectives. Thus neither the general approximation-to-exact idea nor expected polynomial smoothed optimization is new here.

Beier, Röglin, Rösner, and Vöcking [6, Theorem 1] bound smoothed bicriteria Pareto counts with one arbitrary deterministic objective and one linear objective whose independent coefficients have bounded densities and finite first moments. Polynomial moments of smoothed Pareto counts are likewise known [12, 28], and smoothed parametric enumeration has precedent in shortest paths [2, Section 6]. Those results use different perturbation models and outputs. Our general theorem allows arbitrary deterministic Lipschitz branches and adds one shared random linear penalty to their binary labels. To the best of our knowledge, the cited results do not provide the complete bounded-parameter dictionaries with the restricted additive-oracle guarantee proved here. This precise construction and its indicator-quadratic specialization are the principal novelty claims. The companion contributions are the simultaneous restrictions in the hardness and message examples and the two specific star obstructions. To the best of our knowledge, the inspected sources do not contain those restricted statements. Their proofs and comparisons appear with the results; the underlying reduction, pruning, and conic-transfer principles are established.

2 The model and its separator messages

For $n \geq 1$, consider

$$v_\xi = \min \left\{ x^T Q x + c^T x + (\lambda + \xi)^T z : x_i(1 - z_i) = 0 \ (i \in [n]), \ z \in \{0, 1\}^n \right\}. \quad (1)$$

Here $[n] = \{1, \dots, n\}$, $Q \in \mathbb{Q}^{n \times n}$ is symmetric, and $c, \lambda \in \mathbb{Q}^n$. We are given rational bounds

$$0 < \mu I \preceq Q \preceq H I, \quad |c_i| \leq C. \quad (2)$$

Penalties may have either sign. A support means the set $A = \{i : z_i = 1\}$; it need not equal $\{i : x_i \neq 0\}$. In particular, an active coordinate may be zero. All norms below are Euclidean unless specified otherwise.

The support graph of Q has vertices $[n]$ and edges ij for $i \neq j$ with $Q_{ij} \neq 0$. A tree decomposition consists of a tree and vertex sets called bags. Every graph vertex belongs to a bag, every graph edge has both endpoints in a bag, and the bags containing any given vertex form a connected subtree. Its width is the largest bag size minus one. We assume a decomposition of width w with b bags is supplied.

For disjoint sets $I, S \subseteq [n]$ and a box radius $R \geq 0$, define the conditional message

$$M_{I,S}(t) = \min_{\substack{x_I \in \mathbb{R}^I, \ z_I \in \{0,1\}^I \\ x_i(1-z_i)=0 \ (i \in I)}} \left\{ x_I^T Q_{II} x_I + (c_I + 2Q_{IS}t)^T x_I + \sum_{i \in I} (\lambda_i + \xi_i) z_i \right\}, \quad t \in [-R, R]^{|S|}. \quad (3)$$

Boundary unary terms, boundary penalties, and edges within S are excluded. This convention keeps the internal matrix equal to the positive-definite principal matrix Q_{II} .

For $A \subseteq I$, completing the square gives

$$x_A(t) = -\frac{1}{2}Q_{AA}^{-1}(c_A + 2Q_{AS}t), \quad (4)$$

$$q_A(t) = -\frac{1}{4}(c_A + 2Q_{AS}t)^T Q_{AA}^{-1}(c_A + 2Q_{AS}t) + \sum_{i \in A} \lambda_i, \quad (5)$$

$$F_A(t) = q_A(t) + \sum_{i \in A} \xi_i, \quad (6)$$

$$M_{I,S}(t) = \min_{A \subseteq I} F_A(t). \quad (7)$$

The empty support contributes zero. Coordinates outside A are zero. Every q_A is a full quadratic polynomial with rational coefficients, as is F_A when the perturbations are rational; “full” means defined on the whole parameter space, rather than restricted to a region where it is optimal. A dictionary is any subset of the noisy branches F_A whose minimum equals (7) on the specified box.

Root the decomposition. For a child–parent edge, let S be the bag intersection and let I contain the vertices in the child subtree outside S . The running-intersection property implies that no graph edge joins I to a vertex outside $I \cup S$. Indeed, take a bag containing such an edge. If it lies in the child subtree, both endpoints belong to $I \cup S$. If it lies outside, the endpoint in I occurs on both sides of the child–parent edge, forcing it into S . Both cases give a contradiction. Thus (3) is its exact subtree message. We also request $M_{[n],\emptyset}$, whose constant value is the global optimum. These are at most b messages, each with $|S| \leq w + 1$.

Boundary indicators need no separate family of messages. Setting a boundary bit to zero restricts the corresponding coordinate of t to zero. Setting that bit to one permits the same zero value, with its penalty accounted for outside the message. Hence a single dictionary handles all boundary states by restriction to coordinate slices.

3 Counting and finding all relevant supports

The results in this section do not require quadratic functions or a graph. Let Z be a nonempty subset of $\{0, 1\}^m$. The case $m = 0$ consists of one support and follows with the empty-product conventions.

3.1 A near-optimal support count

Assume the independent real random variables ξ_i satisfy

$$\mathbb{P}\{\xi_i \in J\} \leq \phi|J| + \tau \quad \text{for every closed interval } J, \quad \phi > 0, \quad \tau \geq 0. \quad (8)$$

This allows atoms. For arbitrary deterministic costs $g_z \in \mathbb{R}$, put

$$F_z = g_z + \xi^T z, \quad v = \min_{z \in Z} F_z, \quad A_a = \{z \in Z : F_z \leq v + a\}, \quad a \geq 0.$$

Lemma 3.1 (Near-optimal count). *Under (8),*

$$\mathbb{E}|A_a| \leq (1 + 2\phi a + \tau)^m. \quad (9)$$

All tied supports are counted separately.

Proof. A family $A \subseteq \{0, 1\}^m$ *shatters* $D \subseteq [m]$ if its projections onto D contain every binary pattern. Write $\text{Sh}(A)$ for its shattered coordinate sets. The classical sandwich inequality [21, Theorem 5] gives

$$|A| \leq |\text{Sh}(A)|. \quad (10)$$

For completeness, split A on its last coordinate, obtaining projected families A_0, A_1 . Set $U = A_0 \cup A_1$ and $V = A_0 \cap A_1$. Then $|A| = |U| + |V|$. Every set shattered by U is shattered by A , and every set shattered by V extends, by the last coordinate, to a set shattered by A . These two collections are disjoint. Induction proves (10), starting with the empty family, which shatters no set, and the singleton family in dimension zero, which shatters the empty set.

We use the bit-class conditioning method of Beier and Vöcking's generalized isolating lemma [5, conference version, Lemma 5], combined here with (10) and the interval-mass bound. Fix $D \subseteq [m]$ and condition on $(\xi_i)_{i \notin D}$. For a pattern $u \in \{0, 1\}^D$, define

$$C_u = \min_{z \in Z: z_D = u} \left(g_z + \sum_{i \notin D} \xi_i z_i \right).$$

If a class is empty, D cannot be shattered. Otherwise these class costs are finite and independent of ξ_D . If A_a shatters D , every class has a near-optimal representative, so its minimum lies in $[v, v + a]$. Comparing the zero pattern with each unit pattern gives

$$|C_{e_i} + \xi_i - C_0| \leq a \quad (i \in D).$$

These are fixed intervals of length $2a$ for the independent remaining coordinates. Therefore $\mathbb{P}\{D \in \text{Sh}(A_a)\} \leq (2\phi a + \tau)^{|D|}$. Taking expectations in (10) and summing over all D gives (9) by the binomial formula. $\square$

If ξ_i is uniform on the endpoint-inclusive grid

$$\left\{ -\sigma + \frac{2\sigma j}{N-1} : j = 0, \dots, N-1 \right\}, \quad N \geq 2, \quad \sigma > 0, \quad (11)$$

then

$$\phi = \frac{1}{2\sigma}, \quad \tau = \frac{1}{N} \quad (12)$$

are valid. Indeed an interval of length d contains at most $d(N-1)/(2\sigma) + 1$ grid points. This direct estimate includes grid endpoints and exact ties.

3.2 Certified enumeration with an additive oracle

A partial assignment p fixes some coordinates of z ; let Z_p be the corresponding subset of Z . At accuracy $\varepsilon > 0$, assume an oracle either recognizes $Z_p = \emptyset$ or returns $z_p \in Z_p$ and its *exact* cost $V_p = F_{z_p}$ with

$$\min_{z \in Z_p} F_z \leq V_p \leq \min_{z \in Z_p} F_z + \varepsilon. \quad (13)$$

Thus $\ell_p = V_p - \varepsilon$ is a certified lower bound. The oracle must work under arbitrary fixed-bit restrictions.

For a tolerance $\delta \geq 0$, use the following procedure. Query the unrestricted cell and keep its candidate value U fixed. Store nonempty cells in a heap keyed by ℓ_p , caching their candidates.

While the smallest key is at most $U + \delta$, extract that cell and output its candidate z . If its free coordinates in increasing order are $j_1, \dots, j_r$, create the r children

$$z_{j_h} = z_{j_h}^{\text{out}} \ (h < \ell), \quad z_{j_\ell} = 1 - z_{j_\ell}^{\text{out}}, \quad \ell = 1, \dots, r, \quad (14)$$

retaining the parent's fixed coordinates; z^{out} denotes the extracted candidate. Query each child and insert it if nonempty. Stop when the heap is empty or its smallest key exceeds $U + \delta$. This is Lawler's first-difference partition [23], with certified approximate values used in the stopping test.

Lemma 3.2 (Enumeration guarantee). *The procedure terminates, has no repeated outputs, and returns a set E satisfying*

$$A_\delta \subseteq E \subseteq A_{\delta+2\varepsilon}. \quad (15)$$

It uses at most $1 + m|E|$ oracle calls.

Proof. The children in (14) partition the parent cell except for the extracted support: every other support has a unique first differing free coordinate. Hence the stored cells always partition all unextracted supports. Outputs are distinct, so the finite family guarantees termination. Since $v \leq U$, a remaining support of cost at most $v + \delta$ belongs to a cell with lower bound at most its cost and hence at most $U + \delta$. The stopping condition cannot hold while such a support remains. Conversely an extracted candidate has

$$V_p = \ell_p + \varepsilon \leq U + \delta + \varepsilon \leq v + \delta + 2\varepsilon.$$

Each extraction creates at most m child calls, proving the count. $\square$

No exact ranking of the outputs is asserted. The guarantee holds even when each oracle call returns the worst permitted approximate candidate. It is pointwise in the entire noise vector; later probability bounds do not condition on the adaptive search history.

3.3 A complete dictionary on a parameter domain

Let T be a nonempty compact metric space and let deterministic functions $q_z : T \rightarrow \mathbb{R}$ be L -Lipschitz for a common $L \geq 0$. Define

$$F_z(t) = q_z(t) + \xi^T z, \quad V(t) = \min_{z \in Z} F_z(t).$$

The set of supports optimal somewhere is measurable: for each z , its activity is equivalent to $\min_{t \in T} (F_z(t) - V(t)) = 0$, and this minimum is continuous in the finite noise vector. For the expectation statements, the oracle's selected support must be measurable in the noise; all remaining ties are resolved by fixed deterministic orders.

Theorem 3.3 (Complete dictionary from restricted approximation). *Suppose (8) holds, $n \geq \max\{1, m\}$, and $\tau \leq 1/(2n)$. Set*

$$\varepsilon = \delta = \frac{1}{12n\phi}. \quad (16)$$

For $L > 0$, take a deterministic $\varepsilon/(2L)$ -net $t_1, \dots, t_J$ in T . For $L = 0$, take any one point. Run the preceding enumeration at every t_j , using (13) for $F_z(t_j)$, and let $\mathcal{D} = \bigcup_j E_j$. Then, for every noise realization,

$$\bigcup_{t \in T} \arg \min_{z \in Z} F_z(t) \subseteq \mathcal{D}, \quad V(t) = \min_{z \in \mathcal{D}} F_z(t) \quad (t \in T). \quad (17)$$

Moreover,

$$\mathbb{E} \sum_{j=1}^J |E_j| \leq eJ, \quad \mathbb{E}[\text{number of oracle calls}] \leq J(1 + em). \quad (18)$$

Proof. If z minimizes at t and t_j is within $\varepsilon/(2L)$, compare it with a minimizer y at t_j :

$$F_z(t_j) \leq F_z(t) + \varepsilon/2 \leq F_y(t) + \varepsilon/2 \leq F_y(t_j) + \varepsilon.$$

Lemma 3.2 therefore outputs z at t_j . For $L = 0$ the same conclusion holds at the single point because every branch is constant. This proves inclusion, even for supports optimal only at ties or on the boundary. Every extra output is a valid member of Z , which proves equality of envelopes. At each deterministic net point,

$$\mathbb{E}|E_j| \leq \mathbb{E}|A_{3\varepsilon}(t_j)| \leq (1 + 6\phi\varepsilon + \tau)^m \leq (1 + 1/n)^m \leq e.$$

Sum these estimates and use the oracle-call bound. The same random vector is used at every point; independence between points is unnecessary. $\square$

For a rational box $[-R, R]^k$ with $k, R, L > 0$, divide each coordinate into $h = \max\{1, \lceil 2kRL/\varepsilon \rceil\}$ equal intervals and use their product midpoints. The Euclidean covering radius is at most $\sqrt{k}R/h \leq kR/h \leq \varepsilon/(2L)$, and

$$J = h^k \leq (1 + 2kRL/\varepsilon)^k. \quad (19)$$

If $k = 0$ or $R = 0$, the box is a singleton. Rational R, L, ε give rational net points of polynomial encoding length.

To turn the theorem into an algorithm for functions, one also needs efficient construction of each branch representation from its support. Only a first-moment bound is required: at a single net point at most $1 + m2^m$ cells are generated, so every heap operation uses $O(m + \log(m + 1))$ comparisons, even on a large realization. Support strings can be deduplicated in a binary trie in $O(m)$ time per output. Thus the additional work is linear in the output count, up to deterministic polynomial factors. If exact evaluation, branch construction, and each restricted oracle call have uniformly polynomial bit cost, the expected bit cost is polynomial whenever J is polynomial. This conclusion does not require higher moments or construction of optimality regions.

4 Exact messages at fixed treewidth

We now provide the restricted additive oracle and its bit bound for (3). This also fixes the numerical dependence of the main result.

4.1 Uniform conditional bounds

For $k = |S|$ and any $A \subseteq I$, principal submatrices inherit (2), and $\|Q_{AS}\|_2 \leq H$. Thus on $[-R, R]^k$,

$$\|x_A(t)\|_2 \leq \frac{C\sqrt{n} + 2H\sqrt{k}R}{2\mu} \leq M_k := \frac{nC + 2HkR}{2\mu}, \quad (20)$$

$$\|\nabla q_A(t)\|_2 = \|2Q_{SA}x_A(t)\|_2 \leq L_k := 2HM_k. \quad (21)$$

The second formula follows by differentiating (5). The bounds are rational, uniform over all supports, and unchanged when some indicator bits are fixed. The term involving R is needed: a coordinate bound for the global optimizer alone need not bound conditional optimizers with a prescribed boundary.

4.2 A restricted grid oracle

Fix a rational boundary t and an accuracy $\varepsilon > 0$. First suppose $M_k > 0$. Let B be the least positive even integer satisfying

$$B^2\varepsilon \geq HnM_k^2. \quad (22)$$

This choice gives $B \leq 2 + M_k\sqrt{Hn/\varepsilon}$. The square root is used only for this bound; B is found by integer comparisons of the rational inequality. Use the coordinate grid

$$G_B = \{-M_k + 2M_kj/B : 0 \leq j \leq B\}.$$

Each vertex $i \in I$ has an inactive state $(z_i, x_i) = (0, 0)$ and active states $(1, a)$ for $a \in G_B$. Since B is even, active zero is included and is distinct from inactive zero. Fixed bits delete incompatible states.

The grid objective is a sum of the unary costs

$$Q_{ii}x_i^2 + (c_i + 2(Q_{Ist})_i)x_i + (\lambda_i + \xi_i)z_i$$

and the edge costs $2Q_{ij}x_ix_j$ for $i < j$ in I with $Q_{ij} \neq 0$. Restrict every bag of the supplied decomposition to I ; empty bags are harmless. Assign each unary and edge cost to one bag containing its variables, exactly once. For a bag u and an assignment s of its local states, let $a_u(s)$ be its assigned cost. Root this tree, write B_u for its bag, and let $D_u(s)$ be its optimal subtree cost conditional on s . The recurrence is

$$D_u(s) = a_u(s) + \sum_{v \text{ child of } u} \min_{r: r|_{B_u \cap B_v} = s|_{B_u \cap B_v}} D_v(r). \quad (23)$$

The running-intersection property makes different child interiors disjoint and ensures that the separator agreement is sufficient for compatibility. Induction from the leaf bags shows that (23) computes the stated conditional minimum of the factors assigned to each subtree. Minimizing at the root gives the exact grid optimum. Each child table is first minimized over its states at each separator assignment. Reading these projected tables at the parent avoids a product of two bag tables. With at most $B + 2$ states per vertex, the arithmetic operation count is

$$O((b + n^2)(B + 2)^{w+1}). \quad (24)$$

Back-pointers recover the minimizing support. No balance or degree bound on the decomposition is needed.

Lemma 4.1 (Certified additive oracle). *Solve this grid problem under any fixed-bit restrictions. For its support A , return the exact noisy value $F_A(t)$ from (6). This return satisfies (13).*

Proof. Choose a truly optimal support A^* for the restricted continuous problem and its unique continuous minimizer. It lies in $[-M_k, M_k]^I$ by (20). Round active coordinates to G_B and keep inactive coordinates zero. The error e has $\|e\|_2^2 \leq nM_k^2/B^2$. Stationarity on A^* cancels the linear term in this error, so the objective increase is

$$e^T Q_{II} e \leq H\|e\|_2^2 \leq HnM_k^2/B^2 \leq \varepsilon.$$

The optimal grid solution is no worse. Reoptimizing continuously on its support can only decrease its cost, and preserves the fixed bits. Its exact support value is therefore a feasible upper bound within ε of the restricted optimum. $\square$

If $M_k = 0$, every support minimizer is zero. Each free bit then minimizes its noisy penalty $\lambda_i + \xi_i$ independently and fixed bits are retained; this gives an exact oracle directly. Every restriction in this particular model is feasible because an active bit permits zero. The empty-cell convention in Lemma 3.2 is needed for the more general family Z .

4.3 The algorithm and its expected bit complexity

Theorem 4.2 (Complete spectral messages). *Fix w . Given (1)–(2), a width- w decomposition with b bags, rational $R \geq 0$, and rational $\sigma > 0$, let each ξ_i independently be uniform on the grid (11), where N is the smallest power of two at least $2n$. There is an algorithm that, for every noise realization, constructs a rational quadratic dictionary for every rooted subtree message on its box $[-R, R]^{|S|}$ and returns a global optimizer of (1). Each dictionary includes every support optimal anywhere on its box. The algorithm uses $n \log_2 N = O(n \log(n+1))$ random bits.*

More explicitly, put

$$K = w + 1, \quad M = \frac{nC + 2HKR}{2\mu}, \quad L = 2HM, \quad \varepsilon = \frac{\sigma}{6n}, \quad J_* = (1 + 12nKRL/\sigma)^K. \quad (25)$$

Let B be the least positive even integer satisfying (22) with M in place of M_k (take $B = 2$ if $M = 0$). If ℓ is the total binary input length, a polynomial P_w independent of the numerical values bounds the expected number of bit operations by

$$P_w(\ell) bJ_*(B+2)^{w+1}, \quad B+2 \leq 4 + Mn\sqrt{6H/\sigma}. \quad (26)$$

In particular, this is polynomial in the input length and the numerical quantities $C, H, \mu^{-1}, R, \sigma^{-1}$ at fixed w .

Proof. Sample the noise once. For each requested message, use Theorem 3.3 with $\phi = 1/(2\sigma)$, $\tau = 1/N$, and the common bounds M, L . They dominate (20)–(21) for every separator. Use the midpoint net described in (19), or one point for a singleton box or $L = 0$. Its cardinality is at most J_* . At each point the restricted oracle is Lemma 4.1. Form the union of its outputs and construct the noisy branch F_A in (6) for each retained support. Equations (17) and (7) prove correctness for every sample. Minimizing the noisy constants F_A in the empty-boundary dictionary gives a global minimizing support; (4) gives its rational optimizer. Affine minimizer maps may also be stored with every branch.

We detail the bit bound. Every input rational has bit length at most ℓ , and $n, b \leq O(\ell)$. The quantities M, L, ε have polynomial bit length. So do $\log B$, the coordinate subdivision count's logarithm, the net coordinates, and the sampled penalties. This follows directly from their formulas: products, reciprocals of positive rationals, and integer ceilings increase encoding length polynomially. The choice of B uses binary search with rational comparisons and polynomially many bit operations.

Grid unary and edge costs are rational of polynomial bit length. A finite entry in (23) is the value of an assignment to a subset of the original factors, each occurring once. There are only $O(n^2)$ such factors. A common denominator formed from their denominators has polynomial bit length; summing any subset also has polynomial bit length. Taking minima selects values and introduces no new denominators. Thus all additions and comparisons in the grid dynamic program have uniformly polynomial bit cost, including during rare runs with large enumerations.

For exact branch construction, clear denominators in a principal system $Q_{AA}u = d$. The resulting integer matrix and right-hand sides have polynomial bit length. Each determinant is a sum of at most $n!$ products of n such integers, so its bit length is polynomial in ℓ . Since Q_{AA} is nonsingular, rational elimination (or determinant formulas) gives polynomial-size coefficients in (4) and (5). Adding the rational noise penalties gives the same bit bound for F_A . These systems can be solved in polynomial bit time. No numerical conditioning estimate is being used as a substitute for exact arithmetic.

The expected total number of oracle calls is at most $bJ_*(1 + en)$ by (18). The calls have the deterministic cost (24) times a polynomial bit factor. Creating partial assignments, heap operations,

deduplication, exact support evaluations, and branch construction add a polynomial factor per output or call, as established after Theorem 3.3. All these factors are absorbed by $P_w(\ell)$. Linearity of expectation sums the work of the messages even though their interiors overlap and use the same noise. Finally (22) and (25) give the displayed bound on $B + 2$ and hence (26). $\square$

At a rational boundary point, evaluating the retained noisy branches F_A and choosing the smallest returns both the exact value and a conditional optimizer. The theorem gives the message on the entire prescribed box, not only at the net points. The net is used to find the dictionary, not to interpolate its values. It gives no dictionary on an unbounded boundary domain. Exceptional noise samples may require exponentially many outputs; both termination and exactness still hold for them. The expectation concerns running time, not correctness.

4.4 An additive certificate for the original objective

Let v_0 denote the optimum without noise. If $(\hat{x}, \hat{z})$ is the exact perturbed optimizer, define

$$\text{LB} = v_\xi - \sum_i \max\{\xi_i, 0\}, \quad \text{UB} = v_\xi - \xi^T \hat{z}. \quad (27)$$

For every feasible (x, z) , its perturbed value is at least v_ξ and $\xi^T z \leq \sum_i \max\{\xi_i, 0\}$. The unperturbed value is therefore at least LB. Evaluating at $(\hat{x}, \hat{z})$ proves

$$\text{LB} \leq v_0 \leq \text{UB}, \quad \text{UB} - \text{LB} = \sum_i (\max\{\xi_i, 0\} - \xi_i \hat{z}_i) \leq \sum_i |\xi_i| \leq n\sigma. \quad (28)$$

Choosing $\sigma = \eta/n$ gives an always-valid additive gap at most η with polynomial expected work in $1/\eta$ under fixed structural and numerical bounds. This standard perturbation certificate is not a relative approximation guarantee. Exact optimization of the original rational instance does not follow by taking an unspecified infinitesimal noise.

The exact-rational reference `checks/spectral_reference.py` uses the valid message-specific bounds M_k, L_k and mesh in place of their common upper bounds. Independent finite checks include support reoptimization and tie-only labels. This is a constructive complexity result: repeated grid solves and the powers in (26) can be large. Neither a practical solver speedup nor applicability to additional nonlinear constraints follows from the theorem.

5 Exact hardness at width two

Unless $\text{NP} \subseteq \text{ZPP}$, the numerical dependence in Theorem 4.2 cannot in general be replaced by dependence on encoding length. We first give the underlying hardness result. Additive constants in the objectives below are included to display nonnegative residuals; they can be subtracted from the decision threshold.

Theorem 5.1. *Fix a rational $0 < \theta \leq 1/10$, and put*

$$\Delta = \frac{\theta^2(1 - \theta^2)}{1 + \theta^4}.$$

The rational threshold decision problem for indicator quadratic optimization is NP-complete under each of the following restrictions. In every case the matrix has bandwidth two, maximum graph degree at most four, treewidth and pathwidth two, and

$$(1 - 3\theta)I \preceq Q \preceq (1 + 3\theta + 2\theta^2)I. \quad (29)$$

1. The matrix depends only on the dimension and θ , all penalties are positive, and the reduction has value at most $\Delta/4$ on YES instances and greater than Δ on NO instances. Linear coefficients and penalties may have exponentially large magnitudes.
2. The same matrix family has $|c_i| \leq 2$ and $0 < \lambda_i \leq 1$. The decision gap may be exponentially small.
3. All penalties equal one and $|c_i| \leq 9$. The matrix now depends on the input, through possibly exponentially small nonzero entries.

The data have polynomial binary encoding length. No strong NP-hardness claim, or fixed positive gap in the last two cases, is made.

Proof. Take a SUBSET SUM instance [20] with positive integers $a_1, \dots, a_n, B$, where $n \geq 2$. For the first claim define $A_i = a_i \theta^{-(n-i+1)}$, $\eta = \Delta/(4n)$, and $s_0 = 0$. Use indicators z_i, w_i for the respective continuous variables x_i, s_i , and minimize

$$F = \sum_i (x_i^2 - 2A_i x_i + A_i^2 z_i) + \sum_i (s_i - \theta s_{i-1} - \theta x_i)^2 + \theta^2 (s_n - B)^2 + \eta \sum_i w_i. \quad (30)$$

On indicator-feasible points, the first summand is $(x_i - A_i z_i)^2$. This identity is used only on that feasible set; the continuous quadratic matrix in (30) is independent of the indicators. If $\sum_i a_i z_i = B$, take $x_i = A_i z_i$, $s_i = \theta s_{i-1} + \theta x_i$, and every $w_i = 1$. All squares vanish, giving $F = \Delta/4$.

For the converse, write $e_i = x_i - A_i z_i$, $r_i = s_i - \theta s_{i-1} - \theta x_i$, and $u = s_n - B$. For every feasible point,

$$B - \sum_i a_i z_i = \sum_i \theta^{n-i+1} e_i + \sum_i \theta^{n-i} r_i - u.$$

On a NO instance the left side has absolute value at least one. Cauchy-Schwarz, applied to $(e, r, \theta u)$, gives

$$F \geq \left[\theta^{-2} + (1 + \theta^2) \sum_{j=0}^{n-1} \theta^{2j} \right]^{-1} > \Delta.$$

Thus arbitrary continuous adjustments cannot remove the gap.

In the order $(x_1, s_1, \dots, x_n, s_n)$, every diagonal entry is $1 + \theta^2$. The only off-diagonal entries are

$$Q_{x_i, s_i} = -\theta, \quad Q_{s_{i-1}, s_i} = -\theta, \quad Q_{s_{i-1}, x_i} = \theta^2 \quad (i \geq 2), \quad (31)$$

with only the first type present at $i = 1$. The bags $\{x_1, s_1\}, \{s_1, x_2, s_2\}, \dots, \{s_{n-1}, x_n, s_n\}$ give a width-two path decomposition; a triangle makes the width exactly two. The maximum degree is at most four. Absolute off-diagonal row sums are at most $3\theta + \theta^2$, proving (29) by Gershgorin. In particular, the matrix is strictly diagonally dominant. For fixed rational θ , the powers defining A_i and all remaining data have polynomial encoding length. The rational threshold $\Delta/4$ therefore proves NP-hardness. It also proves NP-hardness of approximation within absolute error $\Delta/4$, since the resulting YES and NO intervals are disjoint.

For the second claim let $M = 1 + B + \sum_i A_i$, substitute $(x, s) = M(\hat{x}, \hat{s})$, and divide the objective by M^2 . The quadratic matrix is unchanged, A_i, B become $A_i/M, B/M$, and the state penalty becomes η/M^2 . This gives the stated coefficient bounds and scales both thresholds by M^{-2} . Even the optimizing coordinates are bounded: the all-zero point has value at most θ^2 , so every optimal control and dynamic residual has absolute value at most θ . Hence $|\hat{x}_i| \leq 1 + \theta$ and $|\hat{s}_i| \leq \theta(2 + \theta)/(1 - \theta) < 1$.

For the third claim, restrict to $0 < B \leq A := \sum_i a_i$, since the other instances are trivial. Set

$$b_i = \frac{a_i \theta^i}{A}, \quad T = \frac{B \theta^n}{A}, \quad D = 4,$$

so $0 < b_i \leq \theta$ and $0 < T \leq 1$. With $t_0 = D$, indicators z_i, ζ_i , and no constraints beyond their zero indicators, minimize

$$H = \sum_i (x_i^2 - 2x_i + z_i) + \sum_i [t_i - \theta t_{i-1} - b_i x_i - D(1 - \theta)]^2 + \theta^2 (t_n - D - T)^2 + \sum_i \zeta_i. \quad (32)$$

The first dynamic residual is $t_1 - b_1 x_1 - D$. For each z , let $s_0^z = 0$ and $s_i^z = \theta s_{i-1}^z + b_i z_i$. Then $0 \leq s_i^z \leq \theta^i$ and $s_n^z = T$ exactly when $\sum_i a_i z_i = B$. Put $e = x - z$, $d = t - D\mathbf{1} - s^z$, and let L be the lower shift matrix. The control and dynamic squares are

$$\|e\|^2 + \|(I - \theta L)d - \text{Diag}(b)e\|^2 \geq (1 - 3\theta)(\|e\|^2 + \|d\|^2). \quad (33)$$

Indeed, in this quadratic form a control row has diagonal $1 + b_i^2$ and off-diagonal absolute sum at most $(1 + \theta)b_i$. An internal state row has diagonal $1 + \theta^2$ and row sum at most $2\theta + b_i + \theta b_{i+1} \leq 3\theta + \theta^2$; the final row has diagonal one and row sum at most 2θ . These bounds prove (33).

If k state indicators vanish, the corresponding $d_i \leq -D$, and thus

$$H \geq n - k + (1 - 3\theta)D^2 k \geq n + 10.2k.$$

The feasible choice $x = z = 0$, $t = D\mathbf{1}$, $\zeta = \mathbf{1}$ has value $n + \theta^2 T^2 < n + 1$. Consequently every optimum has all state indicators on. Its value is n plus a sum of squares, and equals n exactly for a YES instance. Positive definiteness gives attainment, so every NO value is strictly greater than n . To quantify the gap, let $r_i = t_i - \theta t_{i-1} - b_i x_i - D(1 - \theta)$ and $u = t_n - D - T$. Using $e = x - z$ and $t_0 = D$, the recurrence gives

$$\frac{\theta^n}{A} \left(B - \sum_i a_i z_i \right) = \sum_i \frac{\theta^n a_i}{A} e_i + \sum_i \theta^{n-i} r_i - u.$$

At a NO optimum, Cauchy-Schwarz therefore gives

$$H - n \geq \frac{\theta^{2n}/A^2}{\theta^{2n} \sum_i a_i^2/A^2 + \sum_{j=0}^{n-1} \theta^{2j} + \theta^{-2}} > 0.$$

The residual vector is $(e, r, \theta u)$, and the numerator can be exponentially small.

The homogeneous matrix has diagonals $1 + b_i^2$ on controls and $1 + \theta^2$ on states. Its edges are $-b_i, -\theta, \theta b_i$ in the positions of (31). All are nonzero, and the same row bounds prove the graph claims and (29). Writing $h_1 = D$, $h_i = D(1 - \theta)$ for $i > 1$, the linear coefficients are $-2 + 2h_i b_i$ on controls, $-2h_i + 2\theta h_{i+1}$ on nonterminal states, and $-2h_n - 2\theta^2(D + T)$ on the last state. Their magnitudes are below nine. The constant is at most $16n + 1/4$ and all data have polynomial encoding length.

Finally, for any rational positive-definite instance, a selected support is an NP certificate. Its unique continuous optimum is $-Q_{JJ}^{-1} c_J/2$ and its value has polynomial rational encoding length, by determinant bounds. Exact linear algebra verifies the threshold in polynomial time. Supports containing zero optimizing coordinates remain valid certificates. This proves membership in NP in all three cases. $\square$

For the first family, dividing the objective by $1 + \theta^2$ gives a unit-diagonal matrix with $\|Q - I\|_2 \leq (3\theta + \theta^2)/(1 + \theta^2) \leq 4\theta$. The third family directly has $\|Q - I\|_2 \leq 3\theta + 2\theta^2$. Thus either hardness construction can approach the identity arbitrarily closely. These proofs do not combine an input-independent matrix with unit penalties. Also, sign changes cannot make the triangle edges all nonpositive: their sign product is positive and is unchanged by coordinate sign changes.

Corollary 5.2. *For any fixed rational noise radius $\sigma > 0$, an exact solver with expected bit complexity polynomial only in the encoded input length, uniformly over all linear coefficients and penalties under the finite-grid noise model, would imply $\text{NP} \subseteq \text{ZPP}$, even for the fixed matrix family in Theorem 5.1(1).*

Proof. There are $n_{\text{tot}} = 2n$ indicators. Choose an integer $s \geq 1$ with $3s^2\Delta/4 > 2n_{\text{tot}}\sigma$ and $s^2\Delta/(4n) > \sigma$; such an s has polynomial encoding length and is computable by rational comparisons. Let c_0 denote the additive constant in the expanded objective (30). Replace (c, λ, c_0) by $(sc, s^2\lambda, s^2c_0)$, keeping Q . The substitution $y = sx$ multiplies every old objective by s^2 . Any noise with $|\xi_i| \leq \sigma$ changes the optimum by at most $n_{\text{tot}}\sigma$. The perturbed YES value is therefore at most $s^2\Delta/4 + n_{\text{tot}}\sigma$, whereas the NO value exceeds $s^2\Delta - n_{\text{tot}}\sigma$. These thresholds are disjoint for every draw, and all perturbed penalties remain positive. Sample the solver’s rational finite-grid noise and compare its exact answer with their midpoint. The assumed expected bound would give a zero-error expected-polynomial algorithm for SUBSET SUM. $\square$

This scaling changes both c and λ . It does not establish the same obstruction under bounded indicator penalties.

Continuous chains encoding SUBSET SUM already occur in conditional linear Gaussian inference [26, Section 3]; that mechanism is not new. The contribution here is the simultaneous restriction to one positive-definite quadratic, uncoupled zero indicators with positive costs, and a near-identity bandwidth-two matrix, with the three distinct numerical regimes above. Das and Kempe’s bandwidth FPTAS [15, Theorem 4.1] constrains the predictor-only covariance graph, the graph corresponding to our quadratic matrix. Their exact tree result [15, Theorem 5.1] instead includes the response vertex. They impose a cardinality constraint, and the FPTAS gives a relative approximation to explained variance; those guarantees differ from our penalized exact problem. General sparse approximation hardness also precedes this work: Çivril [14, Theorem 1.1] proves inapproximability for a cardinality-constrained squared-projection objective, without the graph, positive-definiteness, or spectral restrictions used here. The additive approximation guarantee for banded indicator quadratics in [19, Theorem 1] depends on numerical linear-coefficient and accuracy bounds, consistently with the present reductions.

6 Large exact messages from fixed local data

Hard optimization and large conditional messages are different obstructions. The example below has an easy unconditioned optimum, but every one of its support quadratics is needed in an exact scalar message. This section is deterministic: $\xi = 0$. Its worst-case dictionary lower bound does not contradict the expected bound under independent penalty perturbations in Theorem 4.2. Fix rational $0 < \theta \leq 1/10$ and define, for $t \in [0, 1]$,

$$V_n(t) = \min \left\{ \sum_{i=1}^n (x_i^2 - 2x_i + z_i) + \sum_{i=1}^n (s_i - \theta s_{i-1} - \theta x_i)^2 : \begin{array}{ll} x_i(1 - z_i) = 0, & z_i \in \{0, 1\}, \\ s_0 = 0, & s_n = t \end{array} \right\}. \quad (34)$$

The same stage objective applies at every step. A terminal equality defines a conditional value function, rather than an extra constraint on the unconditioned optimization model.

Lemma 6.1. Let $W_n = \sum_{j=0}^{n-1} \theta^{2j}$. The exact support values in (34) are

$$q_z(t) = \frac{(t - c_z)^2}{D_z}, \quad c_z = \sum_{i=1}^n \theta^{n-i+1} z_i, \quad D_z = W_n + \sum_{i=1}^n \theta^{2(n-i+1)} z_i. \quad (35)$$

Thus $V_n = \min_z q_z$.

Proof. For fixed z , set $e_i = x_i - z_i$ and $r_i = s_i - \theta s_{i-1} - \theta x_i$; inactive controls require $e_i = 0$. The residual objective is $\sum_i (e_i^2 + r_i^2)$ and its only remaining constraint is

$$t - c_z = \sum_i \theta^{n-i+1} e_i + \sum_i \theta^{n-i} r_i.$$

Conversely, any such residuals reconstruct feasible controls and states. Cauchy–Schwarz is attained at $e_i = (t - c_z) \theta^{n-i+1} z_i / D_z$ and $r_i = (t - c_z) \theta^{n-i} / D_z$, proving the formula. $\square$

Theorem 6.2. Every finite exact representation of V_n on $[0, 1]$ as a minimum of polynomials, or as a piecewise polynomial function, uses at least 2^n distinct polynomial formulas. This holds for conditional messages of pure indicator problems with unit penalties on every variable, a fixed finite alphabet of rational local coefficients, and the graph and spectral bounds of Theorem 5.1 (width at most two for all n , and exactly two when $n \geq 2$).

Proof. Put $C_\theta = (1 + \theta^2)/(1 - \theta^2) < 2$. Every D_z lies in $[1, C_\theta]$, and all centers lie in $[0, \theta/(1 - \theta)]$. Their minimum separation is θ^n . Indeed, if two strings first differ at power $k < n$, their center difference has magnitude at least $\theta^k(1 - 2\theta)/(1 - \theta) \geq \theta^n$; strings differing only at power n attain θ^n . On $|t - c_z| \leq \theta^n/3$, intersected with $[0, 1]$, we have

$$q_z(t) \leq \theta^{2n}/9, \quad q_{z'}(t) \geq 4\theta^{2n}/(9C_\theta) > q_z(t) \quad (z' \neq z).$$

Each q_z therefore agrees with V_n on an ordinary open interval. If finitely many representing polynomials cover that interval, one agrees identically with q_z : finitely many nonzero polynomial differences have only finitely many roots. This argument applies both to a minimum and to a piecewise representation, even when the permitted polynomial degrees exceed two. Distinct centers give distinct polynomials.

Before conditioning s_n , the quadratic matrix in (34) has the edges (31), control and nonterminal-state diagonals $1 + \theta^2$, and final diagonal one. The same row bounds prove (29). To add unit indicators to all states, let $y_0 = 4$, attach ζ_i to y_i , and replace the dynamic terms by

$$\sum_i [y_i - \theta y_{i-1} - \theta x_i - 4(1 - \theta)]^2 + \sum_i \zeta_i, \quad y_n = 4 + t. \quad (36)$$

The separator is $S = \{y_n\}$; the lower bound already holds on $y_n \in [4, 5] \subseteq [-R, R]$ for $R \geq 5$, corresponding to $t \in [0, 1]$. For a given z , its reference states satisfy $s_0^z = 0$, $s_i^z = \theta s_{i-1}^z + \theta z_i \geq 0$. With $e = x - z$ and $d = y - 4\mathbf{1} - s^z$, the squares form the same quadratic matrix just bounded. If k state indicators are off, the objective is at least $n - k + 16(1 - 3\theta)k \geq n + 10.2k$. An all-on state choice with $x = z = 0$, $y_i = 4$ for $i < n$, and $y_n = 4 + t$ costs $n + t^2 \leq n + 1$. Hence every minimizing state indicator is on. Before removing boundary-only terms, the conditional value is $n + V_n(t)$; excluding the terminal indicator penalty gives $n - 1 + V_n(t)$. Our message convention also subtracts the remaining boundary-only polynomial. This common subtraction does not change the number of indispensable polynomial formulas. Expanded linear coefficients have magnitude at most eight; the quadratic and linear coefficients and unit penalties belong to a finite alphabet depending only on θ . The removable additive constant is at most $16n$. $\square$

This is a lower bound on explicit formulas, not on every representation or optimization algorithm. Without conditioning the terminal state, every control support has a zero-residual trajectory $x = z$, so the original objective has value zero and the lifted objective has value n .

6.1 Matching bounds for pruning original quadratics

Let $N_n(\varepsilon)$ be the smallest number of the original quadratics q_z whose minimum R satisfies $0 \leq R - V_n \leq \varepsilon$ on $[0, 1]$. Set $N_\theta(\varepsilon) = \sup_{n \geq 1} N_n(\varepsilon)$.

Theorem 6.3. *As $\varepsilon \downarrow 0$,*

$$N_\theta(\varepsilon) = \Theta_\theta(\varepsilon^{-\alpha_\theta}), \quad \alpha_\theta = \frac{\log 2}{2 \log(1/\theta)}.$$

The lower bound concerns pruning original support quadratics only.

Proof. Retain the last m bits of a support, where $0 \leq m \leq n$, and let

$$\delta = \sum_{j=m+1}^n \theta^j \leq \frac{\theta^{m+1}}{1-\theta}, \quad \beta = \sum_{j=m+1}^n \theta^{2j} \leq \frac{\theta^{2m+2}}{1-\theta^2}.$$

For each suffix keep both the all-zero and all-one omitted prefixes. These give at most 2^{m+1} original quadratics, retaining the original W_n in their denominators. For an optimal support, the two retained centers c_0, c_1 bracket its center and have distance δ . The endpoint closest to t has squared distance at most $(t - c_z)^2 + \delta^2/4$: outside $[c_0, c_1]$ it is no farther than c_z , and inside the interval it is within $\delta/2$. Its denominator differs from D_z by at most β ; all denominators are at least one and $(t - c_z)^2 \leq 1$. Consequently

$$0 \leq R - V_n \leq \delta^2/4 + \beta \leq A_\theta \theta^{2m+2}, \quad A_\theta = \frac{1}{4(1-\theta)^2} + \frac{1}{1-\theta^2}. \quad (37)$$

Keeping just the zero prefix also gives the simpler bound $R - V_n \leq 2\delta + \beta$ with 2^m terms, by bounding the numerator change by 2δ . Taking $m = \lceil \log(A_\theta \theta^2/\varepsilon)/(2 \log(1/\theta)) \rceil$, or retaining every support if $n \leq m$, proves the upper bound $4(A_\theta \theta^2/\varepsilon)^{\alpha_\theta}$ for sufficiently small ε .

For the lower bound take $n \geq m \geq 1$. The 2^m centers with zero omitted prefix are separated by at least θ^m , and V_n vanishes at each. A retained quadratic below ε at one such center must have its own center within $\sqrt{C_\theta \varepsilon}$. If $\varepsilon < \theta^{2m}/(4C_\theta)$, it can serve at most one of the 2^m centers. Choose $m = \lfloor \log(1/(8C_\theta \varepsilon))/(2 \log(1/\theta)) \rfloor$. Then $N_\theta(\varepsilon) \geq \frac{1}{2}(8C_\theta \varepsilon)^{-\alpha_\theta}$. $\square$

A related example separates exact complexity from approximation even more strongly. In (34) replace the control coefficient θ in residual i by $b_i = 2^{i-1}\theta^i/(2^n - 1)$. The same residual proof gives centers $c_z = h \sum_i 2^{i-1} z_i$, where $h = \theta^n/(2^n - 1)$, and denominators $D_z = W_n + \sum_i \theta^{2(n-i)} b_i^2 z_i \geq 1$. Every quadratic is again indispensable, but the piecewise function equal to zero on $[0, \theta^n]$ and to q_1 on $[\theta^n, 1]$ underestimates the message by at most $h^2/4$. On the first interval use a nearest center; on the second the all-one support has both the nearest center and the largest denominator. This two-piece function is continuously differentiable at its breakpoint. It is a piecewise approximation, not the pointwise minimum of its two formulas over the entire interval. The single quadratic q_1 also overestimates the message by at most θ^{2n} : its value is at most that bound on $[0, \theta^n]$, and it is exact thereafter.

Exponential growth of piecewise quadratic messages is known for broader nonconvex models [22, Section 5]. Power-law approximation rates for quadratic bases with a common Hessian are established

[18, Section III, Theorem 3.3]; pruning under uniform error also has a covering interpretation [18, Section VI]. Our contribution is the fixed-alphabet, well-conditioned indicator realization and its matching pruning exponent, including its support-dependent denominators. We claim neither a new general phenomenon of message growth nor a lower bound for arbitrary approximating functions.

Multi-period indicator models also admit polynomial algorithms when their projected quadratic matrices are factorizable [25]. Eliminating the free internal states of (34) gives the control matrix

$$\tilde{Q} = I + \frac{uu^T}{W_n}, \quad u_i = \theta^{n-i+1}, \quad \tilde{Q}^{-1} = I - \frac{uu^T}{W_n + u^T u}.$$

Every off-diagonal inverse entry is nonzero. For $n \geq 3$, this natural scalar projection is therefore not factorizable, since a nonsingular factorizable matrix has a tridiagonal inverse [25, Proposition 1]. This comparison does not exclude every possible block reformulation.

A Two representation limits on stars

Exact tree optimization is polynomial [8], but a particular universal representation may still be large. We give two such limits. Neither is a lower bound for arbitrary algorithms or for the original quadratic-indicator epigraph hull.

A.1 A face of the inverse-principal polytope

For positive-definite Q , let

$$P_Q = \text{conv}\{(\mathbf{1}_S, W^S) : S \subseteq \{0, \dots, n\}\}, \quad W_{SS}^S = Q_{SS}^{-1}, \quad W_{ij}^S = 0 \text{ otherwise.}$$

For $S = \emptyset$, set $W^S = 0$. This established polytope occurs in the convexification of Wei et al. [30, Section 3.1 and Theorem 1]. Liu et al. [27, Section 4.3, Proposition 6] already prove computational hardness for a related inverse-graph closure using a diagonal-minus-rank-one matrix. The following result gives an unconditional representation bound for sparse, uniformly conditioned matrices.

Theorem A.1. *For $m \geq 1$, let $b = 1/(2m)$ and*

$$Q = \begin{pmatrix} 1 & b\mathbf{1}^T \\ b\mathbf{1} & I_{2m} \end{pmatrix}.$$

Then Q has a star support graph, rational entries of $O(\log(m+1))$ bits, and condition number below six. The polytope P_Q has an exposed face affinely isomorphic to $\text{COR}(m) = \text{conv}\{aa^T : a \in \{0, 1\}^m\}$. The same conclusion holds for the Stieltjes matrix obtained by changing the sign of the root coordinate.

Proof. The eigenvalues are one and $1 \pm 1/\sqrt{2m}$, so the condition number is at most $(1 + 1/\sqrt{2})/(1 - 1/\sqrt{2}) < 6$. At an atom with root selected, write $z \in \{0, 1\}^{2m}$ for the leaf indicators, $k = \mathbf{1}^T z$, and $t = (1 - b^2 k)^{-1}$. Block inversion gives

$$W_{00} = t, \quad W_{0L} = -btz^T, \quad W_{LL} = \text{Diag}(z) + b^2 t z z^T. \quad (38)$$

At a root-inactive atom the leaf block is simply $\text{Diag}(z)$. Consider the affine functional

$$g(z_0, z, W) = (m+1)(1 - z_0) + m - \mathbf{1}^T z + b^{-2} \sum_{i=1}^m W_{i, m+i}.$$

If the root is inactive, $g = 2m + 1 - k \geq 1$. Otherwise let p be the number of fully selected pairs $(i, m+i)$. Since $k \leq m+p$, we obtain $g = m - k + tp \geq (t-1)p \geq 0$. It vanishes exactly when $p = 0$ and $k = m$: the root is selected and exactly one leaf in each pair is selected. Every other atom has $g \geq b^2$. Indeed, when $p = 0$ its positive value is integral, and when $p > 0$, $(t-1)p = b^2kp/(1-b^2k) \geq b^2$. Thus $g = 0$ exposes precisely the convex hull of atoms with leaf vector $z = (a, \mathbf{1} - a)$, $a \in \{0, 1\}^m$. This atom gap is not an approximate extension-complexity bound.

On this face $t = \tau := (1 - mb^2)^{-1}$ is constant. The affine map $X_{ii} = z_i$ and $X_{ij} = W_{ij}/(b^2\tau)$ for $i \neq j$ maps onto $\text{COR}(m)$. To prove injectivity as well, put $a = \text{diag}(X)$, $z = (a, \mathbf{1} - a)$, and

$$R(X) = \begin{pmatrix} X & a\mathbf{1}^T - X \\ \mathbf{1}a^T - X & \mathbf{1}\mathbf{1}^T - a\mathbf{1}^T - \mathbf{1}a^T + X \end{pmatrix}.$$

Recover $z_0 = 1$, $W_{00} = \tau$, $W_{0L} = -b\tau z^T$, and $W_{LL} = \text{Diag}(z) + b^2\tau R(X)$. At binary atoms these are exactly (38); affinity extends both inverse identities to their convex hulls. Finally, if $D = \text{Diag}(-1, 1, \dots, 1)$, every padded principal inverse for DQD is DW^SD . The invertible linear map $(z, W) \mapsto (z, DWD)$ identifies the two polytopes. $\square$

Corollary A.2. *For the matrices in Theorem A.1, an exact lift of P_Q requires exponentially many scalar nonnegative factors in an LP lift, exponential total cone dimension in a second-order-cone lift, and exponentially many positive-semidefinite blocks of any fixed order. An unrestricted semidefinite lift requires total block order $2^{\Omega(m^{2/13})}$. Scalar inequalities count as order-one factors.*

Proof. Intersect the affine slice of any lift with the equation $g = 0$ and compose its projection with the face isomorphism. This gives a lift of $\text{COR}(m)$ over the same cone; no duality or strict-feasibility argument is needed. Apply the fixed-order block bounds of Fawzi and Parrilo [17, Theorem 1]. In particular, an order-two lift needs at least $7^{-1/2}(9/7)^{m/2}$ blocks. A Lorentz cone of dimension $k \geq 3$ has a lift with $k - 2$ three-dimensional Lorentz cones, each isomorphic to a positive-semidefinite cone of order two. Dimensions one and two need at most their dimension in scalar cones. Hence total second-order-cone dimension D yields at most D order-two blocks, giving the stated exponential bound. LP factors embed in order-two blocks as well. For arbitrary semidefinite blocks, combine them into one block with order equal to their total order and apply Lee, Raghavendra, and Steurer [24, Theorems 1.1 and 5.4]. $\square$

The exponent $2/13$ is a valid established bound, not a claim of the strongest currently known exponent. The SOCP conclusion concerns total dimension, not the number of factors when their dimensions may grow. Also, projecting out inverse entries can remove this obstruction. On the exposed face, all indicators, inverse diagonals, and inverse entries on the star edges depend only affinely on a ; their projection is a cube. The correlation information lies in leaf-leaf inverse entries. Thus this argument gives no lower bound for epigraph formulations that eliminate W .

A.2 Exact projected-row decision diagrams

Choi et al. [13, Definition 7 and Lemma 2] store the future rows of a factor F , where $Q = FF^T$, after orthogonal projection away from selected past rows. Section EC.2.2 of that work also permits normalizing each stored row. Their Section 7.2 already observes exponential size on stars. The following bound applies to every variable order and every admissible choice of nonzero weights, including uniform weights with bounded conditioning, and survives row normalization. It concerns this specific state definition.

Theorem A.3. *Let $n \geq 1$, $Q_{00} = Q_{ii} = 1$, $Q_{0i} = Q_{i0} = b_i \neq 0$ for $1 \leq i \leq n$, and all other off-diagonal entries be zero, where $\sum_i b_i^2 < 1$. For unconstrained binary supports, every variable order in the exact projected-row diagram has a layer with at least 2^{n-1} distinct states. The conclusion persists under arbitrary nonzero rescaling of individual stored rows. In particular, it holds for $b_i = 1/(2n)$, for which $\|Q - I\|_2 = 1/(2\sqrt{n})$.*

Proof. The factor rows $f_0, f_1, \dots, f_n$ are linearly independent because Q is positive definite. Leaf rows are orthonormal and $\langle f_0, f_i \rangle = b_i$. Consider the layer immediately before the last leaf j is processed. All other $n - 1$ leaves are in the past. Vary their selected subset S . If the root is still future, its residual row is

$$r_0(S) = f_0 - \sum_{i \in S} b_i f_i.$$

Distinct subsets give nonproportional vectors: their coefficient on f_0 is always one, so proportionality would imply equality and then equality of every membership bit.

If the root was processed, restrict its indicator to one. First project away from the selected leaves and then away from $r_0(S)$. Writing $d_S = \|r_0(S)\|^2 = 1 - \sum_{i \in S} b_i^2 > 0$, the last leaf row is

$$r_j(S) = f_j - \frac{b_j}{d_S} \left(f_0 - \sum_{i \in S} b_i f_i \right).$$

Its coefficient on f_j is always one. If two such rows were proportional they would be equal. Equality of the f_0 coefficients gives equal d_S , and equality of every other coefficient then gives the same subset, because all b_i are nonzero. Thus in either case the 2^{n-1} subsets yield distinct states even after row rescaling. Normalizing intermediate rows does not change their spans, so it produces scalar multiples of these same residuals. All prefixes are feasible because there are no combinatorial support constraints. The final norm formula follows from the two nonzero eigenvalues $\pm \sqrt{\sum_i b_i^2}$ of $Q - I$. $\square$

Remark A.4 (Unnormalized Gram states and a root-last order). The dyadic choice $b_i = 2^{-i}$ also distinguishes states even if only their *unnormalized residual Gram matrices* are stored. Indeed $d_S = 1 - \sum_{i \in S} 4^{-i}$ is different for every subset S , by uniqueness of base-four expansions with digits zero and one. At the layer just before the last leaf j , if the root is future, its residual Gram diagonal is d_S . If the root has been selected, the displayed residual has squared norm $1 - b_j^2/d_S$, again different for every S . Thus every order has 2^{n-1} distinct unnormalized Gram states for this choice. Its condition number is less than $(1 + 1/\sqrt{3})/(1 - 1/\sqrt{3})$ because $\sum_i 4^{-i} < 1/3$. This Gram conclusion is not asserted after row normalization. Separately, for any nonzero weights in Theorem A.3, if the root is last, varying all n leaves in the first residual formula gives 2^n nonproportional row states immediately before the root.

The ambient directions in these residual rows matter: equal Gram matrices alone do not identify the normalized row states. The theorem does not exclude other exact state definitions, objective-dependent pruning, or approximate merging. It is consistent with the fixed-leaf-count tree bounds of [13, Section 7.2] and with exact polynomial tree optimization [8].

B Higher moments and algebraic consequences

This appendix extends the count used in the main construction to higher moments and derives region and subdivision bounds. The probability argument uses classical shattering and higher-gap methods; the geometric consequences use established bounds for polynomial sign conditions and

cylindrical algebraic decomposition. They do not assert new general smoothing or real-algebraic algorithms.

B.1 All fixed moments, including atomic noise

Retain the notation A_a and the interval-mass assumption (8) from Section 3.

Lemma B.1. *For $m \geq 1$, let D_a be the largest size of a coordinate set shattered by A_a . For $1 \leq r \leq m$ and every real $p \geq 1$,*

$$\mathbb{P}\{D_a \geq r\} \leq \binom{m}{r} (2\phi a + \tau)^r, \quad (39)$$

$$\mathbb{E}|A_a|^p \leq [1 + (m+1)^p (2\phi a + \tau)]^m \leq \exp\{m(m+1)^p (2\phi a + \tau)\}. \quad (40)$$

Proof. The conditional class-minimum argument in Lemma 3.1 bounds the probability that a particular r -set is shattered by $(2\phi a + \tau)^r$. A larger shattered set contains a shattered r -set, so a union bound proves (39). The classical Sauer bound follows here directly from (10): every shattered set has size at most D_a , whence

$$|A_a| \leq \sum_{j=0}^{D_a} \binom{m}{j} \leq (m+1)^{D_a}.$$

For the last inequality, encode a subset by its increasing elements padded with zeros to length D_a ; for $D_a = 0$ both sides are one. Set $b = (m+1)^p$. The integer-valued tail identity gives

$$\mathbb{E}b^{D_a} = 1 + \sum_{r=1}^m (b^r - b^{r-1}) \mathbb{P}(D_a \geq r) \leq \sum_{r=0}^m \binom{m}{r} [b(2\phi a + \tau)]^r.$$

The binomial formula and $1 + x \leq e^x$ prove the claim. $\square$

These are moment bounds for one shared random linear penalty, with arbitrary deterministic support costs. Polynomial moments in smoothed optimization are established [12, 28]; the proof above records the particular interval-mass version needed here.

Theorem B.2 (Moments on a compact parameter domain). *Let q_z be deterministic L -Lipschitz functions on a nonempty compact metric space T , and let K count all labels minimizing $F_z(t) = q_z(t) + \xi^T z$ somewhere on T . Fix real $p \geq 1$ and $n \geq \max\{1, m\}$. If*

$$\tau \leq \frac{1}{2n(n+1)^p}, \quad \delta_p = \frac{1}{4n\phi(n+1)^p}, \quad (41)$$

and T has a deterministic $\delta_p/(2L)$ -net of size J_p for $L > 0$, then $\mathbb{E}K^p \leq eJ_p^p$. For $L = 0$ use one point. On $[-R, R]^k$, with $k \geq 1$ and $R > 0$, one may take

$$J_p \leq [1 + 8kRLn\phi(n+1)^p]^k. \quad (42)$$

For continuous noise with densities at most ϕ , one may instead use $\delta_p = 1/[2n\phi(n+1)^p]$. For the grid (11), (41) holds when $N \geq 2n(n+1)^p$.

Proof. The exponent in (40), at $a = \delta_p$, is at most one. Every active support belongs to $A_{\delta_p}(t_j)$ at some net point, by the three-term Lipschitz comparison in Theorem 3.3. Convexity therefore gives

$$K^p \leq \left(\sum_j |A_{\delta_p}(t_j)| \right)^p \leq J_p^{p-1} \sum_j |A_{\delta_p}(t_j)|^p.$$

Taking expectations proves the bound without independence between net points. The midpoint construction (19), with δ_p in place of ε , proves (42). For $L = 0$ every branch is constant on T ; tied constants are still counted separately. Singleton domains and $m = 0$ are immediate. Compactness proves measurability as before Theorem 3.3. Setting $\tau = 0$ permits the stated larger tolerance. The grid assertion follows from (12). $\square$

If $K_1, \dots, K_s$ are message counts with this common bound, then

$$\mathbb{E} \left(\sum_{j=1}^s K_j \right)^p \leq s^{p-1} \sum_{j=1}^s \mathbb{E} K_j^p \leq e(sJ_p)^p. \quad (43)$$

Overlapping subtrees and shared noise do not create an independence assumption. Taking N to be the smallest admissible power of two needs $O_p(\log(n+1))$ random bits per coordinate. This moment order, and hence the grid, must be fixed before drawing the perturbation.

B.2 A companion affine-rank gap estimate

The following alternative tail estimate uses the classical higher-gap conditioning method of Röglin and Teng [28, Sections 6.1–6.2]. It is useful independently of shattering, but is not needed by the main algorithm. The affine rank of a set is the dimension of its affine hull.

Proposition B.3. *For $1 \leq r \leq m$, independent coordinates with densities at most ϕ satisfy*

$$\mathbb{P}\{|A_a| \geq 2^{r-1} + 1\} \leq \binom{m}{r} 2^{r(r+1)} (2\phi a)^r. \quad (44)$$

Under (8), the same bound holds with the last factor replaced by $(2r!\phi a + \tau)^r$.

Proof. A d -dimensional affine subspace admits an injective projection onto some d coordinates, so it contains at most 2^d cube points. The event in (44) therefore supplies $r+1$ affinely independent supports and r coordinates I on which their differences have full rank. Their patterns $y_0, \dots, y_r \in \{0, 1\}^r$ form at most $2^{r(r+1)}$ ordered lists. Condition outside I and define the class costs C_y as in Lemma 3.1. Each selected class minimum is in $[v, v+a]$, so

$$|(y_j - y_0)^T \xi_I + C_{y_j} - C_{y_0}| \leq a \quad (1 \leq j \leq r).$$

The difference matrix D is a nonsingular integer matrix with entries in $\{-1, 0, 1\}$ and $|\det D| \geq 1$. This event lies in a parallelepiped of volume at most $(2a)^r$, giving the density bound. For atomic noise, each cofactor of D has absolute value at most $(r-1)!$, by its determinant expansion. Every row sum of $|D^{-1}|$ is at most $r!$. The parallelepiped is thus contained in an axis-parallel box of side lengths $2r!a$. Conditional independence and (8) give its probability bound. Union over I and the pattern lists finishes both proofs. $\square$

B.3 Connected regions for quadratic branches

For a quadratic family on $T = [-R, R]^k$, $k \geq 1$, $R > 0$, let $\mathcal{A} = \bigcup_{t \in T} \arg \min_z F_z(t)$ and $K = |\mathcal{A}|$. Let $\mathcal{R}$ count the connected components of the open sets

$$U_z = \{t \in (-R, R)^k : F_z(t) < F_w(t) \text{ for all } w \neq z\}.$$

Thus one label can contribute several regions; a tie-only label contributes none but remains in $\mathcal{A}$.

Corollary B.4. *For each fixed k there is a constant C_k such that, deterministically, $\mathcal{R} \leq C_k K^{k+1}$. Under Theorem B.2 with $p = k + 1$, therefore,*

$$\mathbb{E}\mathcal{R} \leq C_k e J_{k+1}^{k+1}. \quad (45)$$

In particular, the endpoint grid with $N \geq 2n(n+1)^{k+1}$ gives polynomial expected connected-region count in every fixed dimension, including $p = 3$ in the plane.

Proof. Deleting never-active labels preserves the complete argmin set at every point, not just its value. For $z \in \mathcal{A}$, the set U_z is one strict sign condition of the $K - 1$ quadratic differences $F_w - F_z$ and the $2k$ linear box inequalities. The classical bound of Basu, Pollack, and Roy [4, Theorem 5.1] bounds its number of components by $C_k(K + 2k)^k$. Their theorem bounds the total number of components of all realizable sign conditions, so it also bounds this one condition; a ball containing the box supplies their bounded ambient convention. No general position is assumed. An identically zero difference simply makes this strict sign condition empty. Summing over z , and absorbing $2k$ into a dimension-dependent constant since $K \geq 1$, proves the deterministic bound. Apply Theorem B.2 for its expectation. $\square$

Tie-only labels are essential to this reduction. The two planar branches 0 and t_1^2 have two open unique-support regions separated by their tie line. Deleting the touching branch leaves one region. More generally, a bound on labels alone cannot control regions for arbitrary Lipschitz functions, which can cross infinitely often. Polynomial structure makes (45) possible. This coarse algebraic bound permits atomic noise but does not preserve the sharper continuous constants proved in Appendix C.

B.4 Constructing a subdivision with every tied label

A sign-invariant cylindrical algebraic decomposition (CAD) partitions real space into cells on each of which specified polynomials have constant signs. Its exact construction has polynomial bit complexity for fixed dimension and degree [3, Introduction]. We use that standard algorithm without requiring adjacency information or a construction of maximal connected regions.

Corollary B.5. *Under the spectral assumptions of Theorem 4.2, fix the width. There is a finite-grid algorithm that constructs, for every requested message, a CAD subdivision of its closed box and lists all minimizing support labels on every cell. It is exact for every draw and has polynomial expected bit complexity in the same numerical parameters as the main theorem. It uses $O_p(\log(n+1))$ random bits per coordinate, with a fixed moment order p depending on the chosen CAD algorithm and the width.*

Proof. Choose the integer $p \geq 1$ below before sampling, and take N to be the smallest power of two satisfying $N \geq 2n(n+1)^p$ in (11). Rerun Theorem 3.3 at accuracy and enumeration tolerance

$$\varepsilon_p = \delta = \frac{1}{12n\phi(n+1)^p}.$$

On the $\varepsilon_p/(2L)$ -net the emitted set satisfies $E_j \subseteq A_{3\varepsilon_p}(t_j)$, and (40) gives $\mathbb{E}|E_j|^p \leq e$. The union $\mathcal{D}$ contains every active label and consists entirely of genuine support branches, with

$$\mathbb{E}|\mathcal{D}|^p \leq e\hat{J}_p^p, \quad \hat{J}_p \leq [1 + 24kRLn\phi(n+1)^p]^k. \quad (46)$$

Use one point in the constant or zero-dimensional cases.

Construct a sign-invariant CAD for every pairwise difference of dictionary quadratics and every box-boundary polynomial, omitting identically zero polynomials while retaining all their support labels. Keep all cells in the closed box, including cells of smaller dimension. At each cell's exact sample point, identify the minimum and list every label attaining it. The signs certify that this list is constant throughout that cell. The complete-label property of $\mathcal{D}$ proves equality with the argmin set of the original family, even on boundary and tie cells. Extra inactive branches may refine the subdivision, but cannot alter its labels.

Uniform Schur-complement bit bounds were proved in Theorem 4.2. For fixed k , standard exact CAD and the additional output-label work therefore cost at most $P_k(\ell)(1 + |\mathcal{D}|)^{a(k)}$, for a fixed computable exponent $a(k)$. Choose an integer $p \geq \max\{1, a(k)\}$, uniformly over separator dimensions. Convexity and (46) bound the expectation of this work. The restricted grid oracle at the new accuracy still has deterministic polynomial cost in the stated numerical parameters; enumeration has polynomial expected work as before. Sum over the requested messages. Shared noise does not require independent dictionaries. $\square$

The cells in this output need not be maximal connected winning regions. The grid is larger than the $N \geq 2n$ grid sufficient for the main dictionary algorithm. Thus the subdivision conclusion uses the higher-moment theorem, not just a first-moment size bound followed by a polynomial-time algorithm.

C Sharper scalar and planar bounds

The coarse algebraic region bound above works in every fixed dimension and allows atoms. The scalar and planar estimates here give sharper dependence on the smoothing scale under continuous noise. We then transfer formula counts to finite grids. Throughout, the coordinate penalties are independent; branch costs themselves need not be independent.

C.1 Scalar ties and formula intervals

Theorem C.1 (Scalar continuous bound). *Let the deterministic branches q_z , $z \in Z \subseteq \{0, 1\}^m$, be L -Lipschitz on a compact interval of length $T > 0$. If the ξ_i have densities at most ϕ , the number H of interior points with at least two minimizing labels and the number I of connected open intervals of unique minimizing support satisfy*

$$\mathbb{E}H \leq 2m\phi LT, \quad \mathbb{E}I \leq 1 + 2m\phi LT.$$

These statements include isolated touching ties and do not require polynomial branches.

Proof. At a fixed parameter t , let $\Delta(t)$ be the difference between the two smallest distinct-label values, counting ties as zero and using infinity for a one-label family. For each coordinate i , condition on all other noise. The two bit-class minima have the form a_i and $b_i + \xi_i$, with a_i, b_i fixed. If the two best labels differ on i , they attain these two class minima. The conditioning argument of Beier and Vöcking's generalized isolating lemma [5, conference version, Lemma 5] therefore gives

$$\mathbb{P}\{\Delta(t) \leq d\} \leq 2m\phi d \quad (d \geq 0). \quad (47)$$

Empty bit classes contribute nothing.

Partition the parameter interval into r equal cells of length $\eta = T/r$. Almost surely, every endpoint and midpoint in these countably many meshes has a unique winner. If a cell contains a tie at t and z wins at its midpoint s , at least one tied label y differs from z . Then

$$0 \leq F_y(s) - F_z(s) \leq 2L|s - t| \leq L\eta.$$

Thus the number C_r of cells containing a tie has $\mathbb{E}C_r \leq 2m\phi LT$ by (47). Every finite subset of the tie set occupies distinct cells for all sufficiently fine meshes. It follows that $H \leq \liminf_r C_r$, also when H is infinite. Fatou's lemma proves the first bound, and hence almost-sure finiteness. Between tie points the unique winning label is locally constant, and therefore constant on each interval. Endpoint ties have probability zero, giving $I \leq H + 1$. The events are measurable: the gap is continuous in the finite branch-value vector and its minimum over a compact cell is continuous in the noise. $\square$

For quadratic branches define a *formula interval* as a maximal positive-length interval on which the envelope has one fixed polynomial formula. Identify identical polynomials, and do not split an interval at an isolated touching tie when the envelope formula stays the same. Separated appearances of one formula count separately.

Lemma C.2 (Full univariate quadratic envelopes). *The lower envelope of $r \geq 1$ distinct full univariate quadratics has at most $2r - 1$ formula intervals. It can be constructed in $O(r \log(r + 1))$ arithmetic, comparison, and quadratic-root operations.*

Proof. The sequence of interval labels has no adjacent repetition and no subsequence a, b, a, b : four strictly winning witness points would force at least three roots of the nonzero quadratic difference. A sequence on r symbols with these two properties has length at most $2r - 1$. Indeed let its first symbol a occur s times. The blocks after successive occurrences of a have disjoint alphabets, since a shared symbol would give a, b, a, b . At least $s - 1$ blocks are nonempty. Applying induction within each nonempty block gives total length at most $s + 2(r - 1) - (s - 1) = 2r - 1$. This is the elementary order-two Davenport–Schinzel bound. To merge two envelopes, traverse their overlapping intervals. On each overlap the two current quadratics have at most two crossings, so merging is linear in the two input lengths. Divide and conquer, using the proved linear length bound at every level, gives the operation count. These are classical envelope operations, not new algorithmic primitives. $\square$

Theorem C.3 (Scalar atomic bound). *For quadratic L -Lipschitz branches on an interval of length T , and the grid noise (11), the formula-interval count I_f obeys*

$$\mathbb{E}I_f \leq 1 + 2m\phi LT + \frac{4m2^m}{N}, \quad \phi = \frac{1}{2\sigma}. \quad (48)$$

Proof. Add independent continuous jitter: $\eta_i = \xi_i + \delta U_i$, with U_i uniform on $[-1, 1]$ and $\delta > 0$. Conditioning on U_i and translating an interval in (12) shows that

$$\mathbb{P}\{\eta_i \in J\} \leq \phi|J| + 1/N.$$

The variables η_i are independent and have densities; the displayed bound is uniform in δ .

Condition on all but η_i and define

$$h_i(t) = \min_{z_i=0} \left(q_z(t) + \sum_{j \neq i} \eta_j z_j \right) - \min_{z_i=1} \left(q_z(t) + \sum_{j \neq i} \eta_j z_j \right),$$

ignoring an empty class. Each class has at most 2^{m-1} branches. Lemma C.2 bounds the common refinement of the two envelopes by fewer than 2^{m+1} pieces. Splitting each quadratic piece at its stationary point gives at most 2^{m+2} monotone arcs. The function h_i is $2L$ -Lipschitz, so its total variation is at most $2LT$. Constant-arc values and arc endpoint values have conditional probability zero for η_i . Every remaining arc contributes at most one level hit and its image has probability at most its length times ϕ , plus $1/N$. Therefore

$$\mathbb{E}\#\{t : h_i(t) = \eta_i\} \leq 2\phi LT + 2^{m+2}/N.$$

Any optimal tie differs on some bit, whose class minima coincide. Summing over bits bounds the jittered tie count, and hence its unique-support interval count, by the right side of (48).

For a fixedunjittered outcome choose ordered interior witness points, one per formula interval, avoiding the finitely many zeros of nonidentical polynomial differences. At each point the winning polynomial has a positive gap from every different polynomial. For all sufficiently small δ , every jittered winner there belongs to this original polynomial class. Consecutive witnesses belong to different classes and require distinct jittered intervals. Along a fixed sequence $\delta \downarrow 0$, ties at these points have probability zero simultaneously. Thus the unjittered formula-interval count is at most the liminf of the jittered counts. Fatou's lemma proves (48). The one-label case $m = 0$ needs no jitter. $\square$

C.2 Compact planar levels and their curvature

The geometric mechanism below is classical. Bourgain, Korobkov, and Kristensen [11, Theorem 6.1, Corollary 6.2, and their proof] use Hessian variation, coarea, and total curvature to control planar bounded-Hessian levels; Ambrosio and Bertrand [1, Section 5.1] develop measure-valued Hessians in a broader difference-of-convex calculus. We give the finite piecewise-quadratic argument, including its interface terms, to specify the constant needed here.

Lemma C.4 (Compact-level bound). *Let $D = [-M, M]^2$, $M > 0$, with perimeter $P = 8M$. Suppose g_0, g_1 are continuous concave L -Lipschitz functions, each quadratic on a finite semialgebraic partition of D . For almost every y , the level set of $h = g_0 - g_1$ in $\text{int } D$ is a finite union of regular arcs with finitely many corners. If $C(y)$ counts its compact connected components, then*

$$\int_{\mathbb{R}} C(y) dy \leq LP/\pi. \quad (49)$$

Proof. Refine the partitions to a finite smooth semialgebraic stratification of cells, interfaces, and vertices, including the boundary. Exclude the critical values of the restriction of h to every stratum. There are only finitely many such values: each critical locus is semialgebraic with finitely many connected components, and the function is constant on each component. Constant restrictions contribute their constant value. Every remaining level is regular within cells and transverse to interfaces, avoids vertices, and consists of finitely many piecewise smooth curves.

Write $|D^2h|_*$ for the total variation of its matrix-valued distributional Hessian using the nuclear norm (the sum of singular values). On a smooth cell, if v is the unit tangent of a regular level curve, its absolute curvature is $|v^T(D^2h)v|/\|\nabla h\|$. The coarea formula therefore bounds the integral over levels of smooth-arc curvature by $\int \|D^2h\|_*$. To include the corners, choose tangent and normal coordinates at an interface. Continuity gives one-sided gradients (a, b_-) and (a, b_+) with the same tangential derivative. Transversality means $a \neq 0$. The angle α between the two level tangents satisfies

$$|a|\alpha \leq |b_+ - b_-|. \quad (50)$$

Indeed the gradients lie in the same half-plane with fixed nonzero tangent coordinate a ; the derivative of their angle as a function of b has absolute value at most $1/|a|$. Their tangent lines turn through this same angle. One-dimensional coarea along the interface multiplies its corner angle by $|a|$, and (50) bounds the resulting integral by $\int |b_+ - b_-| ds$. This is exactly the nuclear-norm mass of the singular Hessian there: the gradient jump is normal, so its Hessian density is a rank-one matrix with that magnitude. Vertices have been excluded by their values. Thus, with total curvature including corners,

$$\int_{\mathbb{R}} \text{TC}(\{h = y\} \cap \text{int } D) dy \leq |D^2 h|_*(\text{int } D). \quad (51)$$

For concave g , the measure $-D^2 g$ is positive semidefinite. Its nuclear mass equals its negative trace, $-\Delta g$, and is at most LP . To see the last inequality explicitly, use inner parallel squares with generic boundaries. The piecewise smooth divergence theorem, including the gradient jumps, identifies the trace mass with minus the outward normal gradient flux. Its absolute value is at most L times that square's perimeter. Increasing the squares to the interior of D and taking the monotone limit proves the bound. Consequently $|D^2 h|_* \leq |D^2 g_0|_* + |D^2 g_1|_* \leq 2LP$. Every compact regular component is a simple closed curve: local regularity and transverse interface crossings give a one-dimensional manifold without boundary. Its total absolute curvature is at least 2π , including corners, by Fenchel's theorem [29, Theorem 2.4]. Divide (51) by 2π to obtain (49). $\square$

C.3 A planar continuous region theorem

Theorem C.5. *Let q_z , $z \in Z \subseteq \{0, 1\}^m$, be quadratics on $D = [-M, M]^2$, $M > 0$. Suppose one common quadratic p makes every $r_z = q_z - p$ concave with $\|\nabla r_z\| \leq L$ on D . For independent noises with densities at most ϕ , let $\mathcal{R}$ count connected open regions in $\text{int } D$ with a unique minimizing label. With $A = 4M^2$ and $P = 8M$,*

$$\mathbb{E}\mathcal{R} \leq 1 + (1 + 1/\pi)m\phi LP + 96 \binom{m}{2} \phi^2 L^2 A. \quad (52)$$

Proof. Subtract p . For a coordinate i with two nonempty bit classes, condition on all other noise and put

$$g_a(t) = \min_{z_i=a} \left(r_z(t) + \sum_{j \neq i} \xi_j z_j \right), \quad h_i = g_0 - g_1.$$

Each g_a is concave and L -Lipschitz; hence h_i is $2L$ -Lipschitz. The equality $h_i(t) = \xi_i$ means exactly that both classes attain the global minimum. Along each boundary edge, one-dimensional coarea and the conditional density bound give at most $2\phi L$ times the edge length expected hits. Every optimal tie is witnessed by a differing bit, so the number B of boundary tie points satisfies

$$\mathbb{E}B \leq 2m\phi LP. \quad (53)$$

Corners are almost surely untied. Quadratic differences restricted to edges have finitely many critical values (or are constant), so almost surely every boundary pair equality is transverse.

Let J count interior points with at least three minimizing labels. Three distinct binary vectors have affine rank two, and some coordinate pair i, j projects them to three corners of $\{0, 1\}^2$. Condition outside that pair and define its four class minima $C_{ab}(t)$, omitting empty classes. Each is L -Lipschitz. A corner triple has an elbow and two neighbors; equality of their three class values gives $(\xi_i, \xi_j) = H(t)$, where each component of H is a signed difference of two class minima. Its derivative rows have norms at most $2L$ almost everywhere, so $|\det DH| \leq 4L^2$. The area formula

and conditional density at most ϕ^2 bound the expected number of its interior solutions by $4\phi^2 L^2 A$. There are four corner triples per coordinate pair. Therefore

$$\mathbb{E}J \leq 16 \binom{m}{2} \phi^2 L^2 A. \quad (54)$$

Class-equality solutions not globally optimal merely overcount. The image under H of a boundary edge has planar measure zero, so there are almost surely no boundary triple ties.

Let W count compact components of the global tie set in $\text{int } D$ with no triple junction; boundary-to-boundary arcs are excluded. On each such loop the same two labels minimize throughout; a change would create a triple tie. Choose a bit on which those labels differ. The loop is an entire compact component of $\{h_i = \xi_i\}$: a further branch of that level would also be a global tie and would require a triple junction. Lemma C.4, applied conditionally, gives

$$\mathbb{E}W \leq m\phi LP/\pi. \quad (55)$$

It remains to bound the graph formed by the ties. Almost surely every pairwise quadratic zero curve is regular or empty. Indeed its nonconstant part is fixed, whereas its constant shift has a density; a singular level of a quadratic must be its unique critical value, if it has one. A constant difference is almost surely nonzero. Also, almost surely no five labels co-minimize anywhere. Five cube points have affine rank at least three, as in Proposition B.3. Choose four with independent differences and three noise coordinates giving an invertible difference matrix. Conditional on the other noise, their equalities place those three coordinates in the image of a polynomial map from D to $\mathbb{R}^3$. That image has three-dimensional measure zero. There are only finitely many choices, proving the assertion. Four-way ties are not excluded.

Away from J and B , a tie is locally a smooth arc between two labels; there is no isolated two-label tie because its equality curve is regular and all other labels remain strictly higher locally. At a junction at most four labels meet, giving at most six pairwise curves and hence degree at most twelve. Semialgebraicity and (54) give a finite graph almost surely. After separating its junction-free loops, the remaining arcs have at most $12J + B$ ends, counted with multiplicity, and hence number at most $6J + B/2$. Isolated junctions do not separate the square. Adding an arc or closed loop increases the number of complementary regions by at most one; equivalently, Euler's formula gives

$$\mathcal{R} \leq 1 + 6J + B/2 + W.$$

Every complementary region has a fixed unique winner. Substitution of (53)–(55) proves (52). $\square$

Both qualifications in the graph argument matter. The affine binary family $F_z(t) = z_1(t_1 + \xi_1) + z_2(t_2 + \xi_2)$ has a four-way tie when $(-\xi_1, -\xi_2)$ lies inside the square. The pair 0 and $\|t\|^2 - r^2$, $0 < r < M$, has a compact tie loop with no junction; small constant perturbations preserve such a loop. The term W cannot be omitted.

The normalization is available for every family of quadratics whose gradients have norm at most K_0 on D . For a unit vector u , evaluate the affine gradient at Mu and $-Mu$ to obtain $\|\nabla^2 q_z\|_{\text{op}} \leq K_0/M$. Thus $p(t) = K_0\|t\|^2/(2M)$ makes $q_z - p$ concave, with gradient bound $L = (1 + \sqrt{2})K_0$. More directly, the boundary-excluded message branches (5) are already concave. Theorem C.5 applies to a two-dimensional spectral message on $[-R, R]^2$, $R > 0$, by taking the planar half-width $M = R$ and $L = L_2$ from (21), without diagonal dominance.

C.4 Distribution discrepancy and necessary formulas

For polynomial branches on a nonempty open domain $T \subseteq \mathbb{R}^k$, let $U(\eta)$ count labels uniquely minimizing somewhere on T . Let $K_f(\eta)$ count distinct polynomial formulas equal to the envelope on a nonempty open subset, identifying identical polynomials. These are the necessary formulas. They still represent the envelope everywhere: outside the finite union of zero sets of nonidentical differences, a winning formula is unique; that set is dense, and continuity supplies the remaining points. On a box, the same reasoning extends coverage to its closed boundary.

Theorem C.6 (Monotone discrepancy transfer). *Let $\mu = \prod_i \mu_i$ and $\nu = \prod_i \nu_i$. Suppose each pair of marginals differs by at most Δ_i on every half-line, with either open or closed endpoint convention. Then*

$$|\mathbb{E}_\nu U - \mathbb{E}_\mu U| \leq |Z| \sum_i \Delta_i. \quad (56)$$

If $\mathbb{E}_\mu U_{q+c} \leq B$ uniformly over deterministic constant branch shifts c , then

$$\mathbb{E}_\nu K_f \leq B + |Z| \sum_i \Delta_i. \quad (57)$$

If instead, for a positive integer p , the uniform bound is $\mathbb{E}_\mu U_{q+c}^p \leq B_p$, then

$$\mathbb{E}_\nu K_f^p \leq B_p + 2|Z|^p \sum_i \Delta_i. \quad (58)$$

Proof. Fix z and all noise coordinates except i . If $z_i = 0$, its unique activity persists as η_i increases: same-bit comparisons stay fixed and opposite-bit comparisons improve. Thus the section of its activity event is an upward half-line, possibly empty or the whole line. For $z_i = 1$ it is a downward half-line. Activity is measurable, since a strict witness can be chosen from a countable dense subset of T . Replace the marginals one at a time, condition on the others, and telescope. The change in the probability of this label's activity is at most $\sum_i \Delta_i$. Summing over labels proves (56). This argument uses only continuous branches.

To handle persistent duplicate polynomials, choose distinct fixed numbers a_z and consider $q_z + \delta a_z$ as $\delta \downarrow 0$. For each necessary original formula, choose a witness in its open winning set outside all zero sets of nonidentical differences. It has a positive gap from every different polynomial. In its identical-formula class, the label with smallest a_z becomes the unique winner at that witness for all sufficiently small $\delta > 0$. There are finitely many classes, so, pointwise in η ,

$$K_f(\eta) \leq \liminf_{\delta \downarrow 0} U_{q+\delta a}(\eta).$$

Fatou's lemma along $\delta = 1/j$ and (56) prove (57). The shifts are a proof device only.

For the last assertion expand U^p over ordered p -tuples of labels, allowing repetitions. A coordinate section of their simultaneous activity is an intersection of upward and downward half-lines, hence an interval. Its discrepancy is at most $2\Delta_i$. The same marginal replacement bounds the difference of the p th moments by $2|Z|^p \sum_i \Delta_i$. Apply the preceding pointwise limit, raised to p , and Fatou's lemma to obtain (58). $\square$

For uniform continuous noise on $[-\sigma, \sigma]$ and its endpoint-inclusive N -point grid, $\Delta_i \leq 1/N$. After rescaling to $[0, 1]$, the grid distribution function is constant between points $j/(N-1)$; comparison with the identity at both ends of every step proves the bound, including open endpoints. Consequently

$$\mathbb{E}_{\text{grid}} K_f \leq B + m2^m/N.$$

Constant shifts preserve the assumptions of Theorems C.1 and C.5, and U is bounded by their region counts. Thus one may use either

$$B = 1 + 2m\phi LT \quad \text{or} \quad B = 1 + (1 + 1/\pi)m\phi LP + 96 \binom{m}{2} \phi^2 L^2 A.$$

For transfer error at most $\epsilon > 0$, a power-of-two grid with $N \geq \max\{2, m2^m/\epsilon\}$ uses $O(m + \log(m+1) + \log_+(1/\epsilon))$ bits per coordinate. Its cardinality is exponential, although its samples have polynomial bit length. These sharper constants concern distinct necessary formulas, not all active labels or connected regions. In particular the scalar correction $m2^m/N$ does not replace $4m2^m/N$ in the formula-interval theorem. The integer-moment transfer has error at most $2m2^{mp}/N$; it requires a continuous p th-moment bound, which the planar first-moment theorem alone does not supply. The direct moment bounds in Appendix B use much smaller grids when sharp first-moment constants are unnecessary.

D Direct recursions under diagonal dominance

The main spectral construction solves each requested message directly. The alternative algorithms here compose child messages. Their stronger numerical assumption gives an invariant box, which makes unrestricted support elimination compatible with recursively retained formulas. Neither algorithm needs bounded degree or bounded branching. Parametric message recursion and support pruning on tree decompositions are established [7, 8]. The particular expected bounds below combine the invariant box with penalty-noise count and moment bounds.

D.1 An invariant box

Assume, in addition to symmetry,

$$0 < d \leq Q_{ii} \leq D_0, \quad \sum_{j \neq i} |Q_{ij}| \leq \rho Q_{ii}, \quad 0 \leq \rho < 1, \quad |c_i| \leq C. \quad (59)$$

For bit-complexity statements, the supplied bounds d, D_0, ρ, C are rational and their encodings are included in the input length ℓ . Strict diagonal dominance makes Q positive definite. If $C > 0$, set

$$M = \frac{C}{2d(1-\rho)}, \quad I = [-M, M]. \quad (60)$$

If $C = 0$, every unrestricted support optimizer is zero and selecting the negative noisy penalties solves the global problem directly; the prescribed invariant box is the singleton zero.

Lemma D.1. *For every fixed support, every conditional continuous optimizer whose boundary coordinates lie in I has all its free coordinates in I . A boundary-excluded branch with k boundary coordinates satisfies, on I^k ,*

$$|\partial q / \partial t_s| \leq 2\rho D_0 M, \quad \|\nabla q\| \leq 2\sqrt{k}\rho D_0 M.$$

Proof. If there is a free active coordinate, let X be their maximum absolute value. Its stationarity equation gives $X \leq C/(2d) + \rho \max\{X, M\}$. If $X > M$, it follows that $(1-\rho)X \leq C/(2d) = (1-\rho)M$, a contradiction. Inactive coordinates are zero; deleting unused edges only decreases the off-diagonal row sums. Differentiating the exact conditional optimum gives $\partial q / \partial t_s = 2 \sum_{i \text{ internal}} Q_{si} x_i(t)$. The invariant box and (59) give the displayed bounds. $\square$

The assertion holds for every support, even one that never wins. This is stronger than a bound on the global optimizer alone. Penalties do not enter the stationarity equations, so the box is uniform over all noise outcomes.

D.2 Bounded-block scalar recursion

Suppose every biconnected block of the support graph has at most b_0 vertices, counting a bridge as a two-vertex block. Blocks need not be cliques. Root each component's block-cut tree at a vertex. Assign each edge cross term to its unique block, and each diagonal, linear, and penalty term to its vertex. A block message $H_B(t)$ excludes its parent vertex's own costs. Represent a scalar envelope by distinct full quadratic formulas necessary on open subintervals of I , with one support representative each and an ordered list of its formula intervals. Deduplicate identical formulas; isolated touches need no separate formula. Continuity preserves values on the whole closed interval.

Write $p_u = \lambda_u + \xi_u$. The active vertex message and inactive atom are

$$F_u(t) = Q_{uu}t^2 + c_u t + p_u + \sum_{B \text{ child of } u} H_B(t), \quad (61)$$

$$A_u = F_u(0) - p_u. \quad (62)$$

For (61), merge child breakpoint lists while maintaining the running sum of their current quadratic coefficients; update only the children that change at a breakpoint. This avoids repeatedly summing all children at a high-degree vertex. Each resulting polynomial is the full conditional value of a genuine combination of child supports. For (62), store one best descendant representative at zero. Switching the vertex bit to zero removes precisely p_u , and descendants interact with the rest only through this fixed zero coordinate. Active zero remains allowed, as required by the original model.

At a block B with parent vertex v , choose for each $u \in B \setminus \{v\}$ either a retained active quadratic or the inactive atom. Add the block's cross terms, set inactive coordinates to zero, and minimize unrestrictedly over the active block coordinates. The resulting Schur complement is a full quadratic in x_v . Construct and prune the lower envelope of these candidates using Lemma C.2.

Theorem D.2 (Block recursion). *Under (59), the preceding recursion constructs exact scalar messages and an optimizer for every penalty vector. With independent noise densities at most ϕ , its expected arithmetic, comparison, and quadratic-root operation count is*

$$O_{b_0} \left(n^2 (2 + 8n\phi\rho D_0 M^2)^{b_0-1} \right). \quad (63)$$

For rational data and the endpoint grid with $N \geq 4n2^n$, its expected bit cost is a factor polynomial in the rational input length and $\log N$ times

$$O_{b_0} \left(n^2 (3 + 8n\phi\rho D_0 M^2)^{b_0-1} \right), \quad \phi = 1/(2\sigma). \quad (64)$$

A smallest admissible power-of-two grid uses $O(n + \log(n+1))$ random bits per coordinate. The continuous statement is an arithmetic-model result, not a bit-complexity claim for arbitrary real samples.

Proof. Every vertex polynomial and block candidate is the exact unrestricted conditional value of a genuine complete support. Positive definiteness makes each eliminated matrix, a Schur complement of a principal matrix, positive definite. Thus no candidate can fall below the true conditional value anywhere, even outside the intervals that led to its construction. Conversely fix a true conditional optimum with parent value in I . Lemma D.1 puts its child boundary values in I . Replace child

descendants by retained representatives attaining their envelopes there, using the inactive atom when appropriate. These replacements preserve value and feasibility. The enumerated tuple then has the true value at those block coordinates, and its unrestricted minimum is no larger. Candidate validity gives the reverse inequality. This proves block coverage; the vertex step follows directly from the sum in (61). Open-interval coverage extends to breakpoints and endpoints by continuity. Induction proves all messages. At a root, compare its inactive atom with the minima of its active quadratics on I ; the invariant box contains every support optimizer. Stored choices and linear support solves recover an optimizer.

Let K_u count retained active formulas at u . The candidate count at block B is at most

$$N_B = \prod_{u \in B \setminus \{v\}} (K_u + 1). \quad (65)$$

After subtracting the common vertex terms, its active support branches have Lipschitz constant $L = 2\rho D_0 M$ on length $T = 2M$, by Lemma D.1. Apply the scalar bounds to the full deterministic support family, not to a random pruned family. Under continuous noise, distinct descendant supports give different noisy polynomials almost surely: at any fixed parameter, their difference contains ξ_i with coefficient 1 or -1 for a bit on which the supports differ. Conditioning on the remaining coordinates makes equality a null event, and there are finitely many pairs. Thus the necessary-formula count is bounded by the scalar unique-support interval count. Theorem C.1 gives $\mathbb{E}(K_u + 1) \leq 2 + 8n\phi\rho D_0 M^2$. For grid noise, Theorem C.3 and $4m2^m/N \leq 1$ for every subtree size $m \leq n$ give $\mathbb{E}(K_u + 1) \leq 3 + 8n\phi\rho D_0 M^2$. Sibling vertex subtrees within one block have disjoint noise coordinates, so their counts in (65) are independent. Fix all representative-selection rules deterministically. This proves the corresponding $(b_0 - 1)$ st power bound on $\mathbb{E}N_B$. Ancestor and descendant counts need not be independent.

There are $O(n)$ blocks. Each candidate requires bounded-dimensional Schur elimination. Every formula has a support representative, hence each list has at most 2^n entries and all logarithmic sorting factors are $O_{b_0}(n)$. Lemma C.2 bounds each block envelope by $2N_B - 1$ pieces. All vertex merges process at most $2\sum_B N_B$ such child pieces. Their sorting, envelope construction, and support bookkeeping therefore fit the coarse deterministic bound

$$O_{b_0}\left(n\left(n + \sum_B N_B\right)\right).$$

Taking expectations proves (63) and the operation part of (64); only first moments and sibling independence were used.

For the bit statement, every stored full polynomial is an original fixed-support Schur complement with rational coefficients of uniformly polynomial bit length, as in Theorem 4.2. The same is true of partial vertex sums, since they include original factors only once. Breakpoints are roots of differences of two such quadratics; they have algebraic degree at most two and polynomial-size defining coefficients. Exact sorting, comparison, sign testing, and root operations thus have polynomial bit cost. Algebraic breakpoints only locate intervals; stored coefficients remain rational, so recursion creates no tower of algebraic extensions. Rational support recovery has the same polynomial bit bound. $\square$

The theorem gives explicit block estimates under stronger hypotheses than the spectral theorem. For $b_0 = 2$, the graph is a forest; Bhathena et al. [8] already solve the unperturbed problem on arbitrary trees in quadratic arithmetic time and memory. The present smoothed bound under diagonal dominance is no improvement in that case. The perturbation certificate (27)–(28) applies unchanged.

D.3 Simultaneous arrangements at fixed treewidth

Bounded block size can be dropped under the same diagonal-dominance assumption. The price is a width-dependent algebraic exponent and a higher-moment argument instead of the explicit scalar operation bound.

Root a width- w decomposition after contracting adjacent bags when one contains the other. There are at most n nonempty bags: every nonroot bag introduces a vertex absent from its parent, and running intersection assigns each vertex to a unique highest bag. Every separator has size at most w . Assign each objective term to its highest containing bag, as in (3). The containing bags of an edge form a connected subtree, so this owner is unique. Each subtree contains all terms involving its internal vertices and no terms involving only its parent separator.

For each boundary indicator mode, fix inactive coordinates to zero. Its $k \leq w$ active coordinates range over I^k ; active zero is allowed. Retain only distinct full support quadratics that uniquely minimize on some relative open subset of this box, after merging identical polynomials. Store one support representative per formula. For $k = 0$, retain one minimum constant. This pruning preserves the envelope on the closed box: outside pairwise equality sets the minimum formula is unique, and such points are dense in the relative interior. A subsequence of their winners is constant when approaching any box point, so continuity gives coverage there.

Theorem D.3 (Direct fixed-width recursion). *Under (59), there is an exact recursive algorithm with deterministic bit work*

$$P_w(\ell)(1 + n + \mathcal{K})^{a(w)}, \quad (66)$$

where $\mathcal{K}$ is the sum of its final dictionary sizes over all bags and boundary modes, P_w is a polynomial, and $a(w)$ is a fixed computable constant. Choose an integer $p \geq \max\{1, a(w)\}$ before sampling, and let N be the smallest power of two with $N \geq 2n(n+1)^p$. Under the grid (11), expected bit work is polynomial in input length and the numerical parameters d^{-1} , D_0 , C , $(1-\rho)^{-1}$, and σ^{-1} at fixed w . The algorithm constructs every mode's formula envelope on its closed box and a global optimizer, correctly for every draw, with $O_w(\log(n+1))$ random bits per coordinate.

Proof. The case $w = 0$ is separable, so assume $w \geq 1$ and $C > 0$. Suppose child dictionaries have been constructed, and fix a bag indicator mode. The active bag coordinates have dimension at most $w + 1$. Lift each child's appropriate dictionary to these coordinates, and form all pairwise quadratic differences *within each child*, together with box boundaries. There are $O_w(1 + \sum_j K_j^2)$ polynomials. Remove identically zero differences and use a fixed rule for identical restricted polynomials. A sign-invariant CAD in this fixed dimension has polynomial size and bit cost [3, Introduction]. On each full-dimensional cell in the relative open box, its sample point selects one minimum formula per child. The selected tuple is minimizing throughout the cell. Thus a simultaneous arrangement discovers the required tuples without a Cartesian product over arbitrarily many children. In dimension zero, select a tuple at the one point.

For each discovered tuple, add its full child polynomials and the owned bag terms. Set inactive variables to zero and unrestrictedly minimize over the active bag coordinates forgotten from the parent separator. Do not impose the cell as an optimization constraint. The resulting full quadratic is the exact conditional value of a genuine internal support: child interiors are disjoint with no cross edges, and elimination is a positive-definite Schur complement. In particular this candidate is everywhere at least the true message value, even if its optimizer leaves its discovery cell.

For the reverse inequality, fix a parent boundary point and an optimal conditional support. Its bag coordinates y lie in the invariant box. Every child envelope is covered there, including at switching surfaces. Approach y by full-dimensional arrangement cells in the bag's relative box. The parent coordinates may vary in this limit as well. There are finitely many tuples, so some subsequence

uses one fixed tuple. Continuity makes its child polynomials attain all child envelope values at y . At y their sum with the owned terms equals the true conditional optimum. Unrestricted elimination of forgotten coordinates yields a candidate no larger than that optimum at the fixed parent boundary point. Candidate validity proves equality. Dimension zero requires only direct evaluation. This argument covers boundary points, lower-dimensional ties, and optima that themselves win on no open cell.

Repeat over the 2^{w+1} bag modes and group candidates by parent boundary mode. A second CAD on each candidate family retains just its uniquely winning distinct polynomials on relative full-dimensional cells. The density and continuity argument preceding the theorem proves that pruning preserves all values on the closed box. At the root, one minimum constant gives the optimum; its support linear system recovers a rational optimizer.

Every local CAD, candidate list, and pruning operation has work polynomial in $1 + \sum_j K_j$ for fixed w . Summing over at most n bags and at most 2^w outgoing modes proves (66) for a fixed computable exponent. Uniform coefficient bit lengths follow from genuine-support Schur complements of the original input. Algebraic CAD sample coordinates select labels only: they are never substituted into the stored coefficients. Thus the recursion retains rational coefficients and the uniform bit bound. This also explains why no recursively growing algebraic extension is hidden in (66).

For the expectation, each final dictionary is bounded by the all-active-label count of its full support family. Lemma D.1 supplies a common Lipschitz bound $L_w = 2w\rho D_0 M$, an upper bound on $2\sqrt{k}\rho D_0 M$ for $k \leq w$. Theorem B.2 gives each dictionary's p th moment at most eJ^p , where

$$J = [1 + 8wML_w n\phi(n+1)^p]^w, \quad \phi = 1/(2\sigma).$$

There are at most $2^w n$ dictionaries. Applying convexity to these counts and $1 + n$ deterministic unit terms gives

$$\mathbb{E}(1 + n + \mathcal{K})^p \leq e[(1 + n + 2^w n)J]^p.$$

Since $p \geq a(w)$, this bounds the expectation of (66). No independence among message dictionaries is used. The bit count follows from the chosen power-of-two grid. $\square$

This algorithm returns formula envelopes and one representative support at each value, rather than every tied label as in Theorem 4.2 and Corollary B.5. Its polynomial degree depends on w ; no fixed-parameter bound with width-independent degree, or practical performance advantage, is claimed. The standard CAD machinery and exact support elimination are classical. The stated expectation follows from their combination with the interval-mass moment bound, not from the planar first moment alone.

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